

Sacrificial failure and minimum-mass design of anchored nets for arresting small drones: impact regime, quasi-static capacity, and a dynamic localization criterion

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(Dated: Working draft and reference skeleton — September 28, 2026)

Spider orb webs survive impacts by allowing heavily loaded threads to fail while the rest of the web remains intact—a behavior often attributed to the strain-stiffening response of silk [1]. We investigate whether this *sacrificial failure* mechanism can minimize the structural mass of anchored, prestressed nets designed to arrest small drones (0.25–2 kg, 10–20 m/s). Extending the transverse-impact theory of Smith *et al.* [3] to a cubic constitutive law $T(\epsilon) = E_0 A \epsilon + \beta \epsilon^3$, we first demonstrate that drone impacts operate deep within a quasi-static regime: the initial impact strain is below 1% of the failure strain, and stress waves traverse the net dozens of times during the arrest process. In this regime, the theoretical minimum net mass simplifies to $m_{\min} = E_{\text{kin}}/(\eta e_{\text{mat}})$, where e_{mat} is the material’s specific energy to failure (J/kg) and $\eta \in (0, 1]$ is the material utilization factor. For a hub impact on a radial net, we derive $\eta = 1 - U(\epsilon_p)/U(\epsilon_b)$, where $U(\epsilon)$ is the strain energy density function, ϵ_p is the prestrain, and ϵ_b is the breaking strain, showing that η is remarkably independent of the number and length of radial threads. We further establish: (i) the transmission and reflection coefficients of tension waves at an N -thread junction, (ii) the dynamic overload factor on adjacent threads following a sudden rupture, and (iii) a dimensionless criterion separating localized sacrificial failure from catastrophic cascading collapse. Explicit mass–spring dynamic simulations confirm these analytical criteria for both a silk-like stiffening material and a Dyneema-like linear elastic material. For impacts in the central half of the net, allowing localized sacrificial failure halves the net mass needed to capture the drone with the silk-like material (factor 2.0–2.1) and saves at most 5% with the linear one, whose stored energy is released into a lateral sling of the drone. Towards the frame the mass needed without failure grows threefold for the silk-like material. In the undamped model a net at the minimum capture mass usually loses further threads on the rebound, so retaining the drone requires dissipation that is not modeled here. Finally, we show that spectral descriptors derived from the net’s force-density Laplacian predict material utilization η , and we train a graph neural network surrogate that accurately predicts thread failure patterns across unseen net topologies with [Z]% accuracy.

I. INTRODUCTION

Nets are one of the few non-destructive means to stop small unmanned aerial vehicles: they can be fired from the ground, carried by an interceptor drone, or installed as fixed barriers around sensitive sites [20]. For fixed or anchored barriers, the design question is simple to state: *what is the lightest net that stops a given drone, wherever it hits?* The answer depends on the material, on the net topology, and on how failure is allowed to proceed.

Spider orb webs solve a closely related problem. Cranford *et al.* showed with molecular-dynamics simulations and field experiments that, under local loading, failure remains confined to the loaded thread, and attributed this to the softening–stiffening response of dragline silk [1]. In their Supplementary Information, a quantized-fracture-mechanics argument [6] predicts the number of broken elements for power-law materials, in quasi-static loading. Sensenig *et al.* measured that radial threads absorb most of the kinetic energy of prey [4], and Aoyanagi and Okumura proposed an analytical model of web mechanics [5]. The transverse impact of a single yarn is classical [2, 3], and extensions to fabrics exist [7–9].

Explicit finite-element models of orb webs have since resolved the dynamics of prey capture: larger webs absorb more kinetic energy before breaking [24], the response to prey impact has been computed with nonlinear silk laws [25], and in oblique impacts damage stays localized mainly through the sacrificial failure of radial threads near the impact point [26]. The architecture of the web has also been treated as a design variable: aerodynamic drag on the web

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changes its optimal topology [19], and experiments on 3D-printed synthetic webs showed that a uniform material distribution is best under localized loading, whereas stronger radials and weaker spirals are best under distributed loading [18].

Engineered nets that arrest impacting bodies have been studied mostly for rockfall protection. Full-scale tests and explicit finite-element models of steel-net barriers, with large deformations, nonlinear net elements, and energy-dissipating brakes, are used for design [27], and simplified energy-momentum models show that the mass of the net controls its out-of-plane displacement more than its stiffness [28]. For drones, fixed nets are reviewed as one of several passive countermeasures [20]; in practice the choice ranges from light nets that snare exposed rotors to heavy, highly tensioned meshes for larger loitering munitions [29]. We did not find peer-reviewed work that sizes an anchored anti-drone net from first principles, and in none of the rockfall or drone-net studies above is the controlled failure of individual net elements used as a design mechanism.

The sudden loss of a load-carrying element is well studied in two other communities. In fiber composites, Hedgepeth computed dynamic overload factors of 1.15–1.27 on the fibers adjacent to a break [10]; in cable-supported bridges, design codes prescribe dynamic amplification factors of 1.5–2.0 for sudden cable loss [11], and the factor depends on how fast the member fails relative to the structural period [12]. Fiber-bundle models describe failure cascades on networks, but under slowly increasing load [13].

In composites, three-dimensional finite-element analyses confirm that dynamic stress concentrations around a fiber break exceed the static ones and extend over a larger region [30]; in cable-stayed bridges and cable-supported structures, nonlinear time-history analyses of cable loss give structure-specific amplification factors [31, 32]. These studies assume linear elastic (or elastoplastic) members and a prescribed break, rather than a break caused by the impact itself. Conversely, the orb-web simulations of Refs. [25, 26] include the nonlinear silk and the impact, but report localization as a numerical observation, without a criterion that predicts it or a measure of the material it saves. What is still missing, to our knowledge, is a criterion that combines (i) the dynamic load transfer at a sudden thread break, (ii) a nonlinear (stiffening) constitutive law, and (iii) the discrete geometry of an anchored net, and that quantifies how much material sacrificial failure can save.

Tavi’s note: Gap check done on 28 Sep 2026 with targeted searches (dynamic amplification in cable nets and cable domes; sudden fiber break with nonlinear fibers; dynamic fiber-bundle models; orb-web impact simulations; rockfall and drone nets). Closest work: Li and Ramakrishnan 2025 [26] (dynamic sacrificial failure of radials, FE, no criterion) and the quasi-static QFM of Cranford SI Sec. S10. The search was not exhaustive; a Scopus/Web of Science pass on “dynamic” + “cable net” + “rupture” is still advisable.

On the computational side, learned simulators based on graph neural networks reproduce the dynamics of particles and meshes at a fraction of the cost [22, 23], and generative models have been used to design spider-web-inspired structures [21]. The force-density method [15, 16] links the equilibrium geometry of a prestressed net to a weighted graph Laplacian, which makes spectral descriptors a natural, compact representation of net designs; topology optimization of force densities has recently been proposed for cable structures [17]. Graph neural networks have been trained to predict fracture: recurrent graph convolutional networks forecast crack evolution and time to failure in brittle material models [33], and graph networks trained on extended finite-element data decide which microcracks propagate and simulate their coalescence [34]. In graph generation, conditioning on the leading eigenvalues and eigenvectors of the graph Laplacian overcomes expressivity limits of one-shot generators [35], and diffusion directly on the truncated spectrum generates graphs at reduced cost [36]. These results suggest that a small set of spectral quantities of the force-density Laplacian may suffice to rank net designs, a hypothesis we test in Sec. VII.

The contributions of this work are:

1. the Smith theory of transverse impact generalized to the cubic law of Eq. (2), and a regime analysis showing that drone impacts on nets are quasi-static;
2. a closed-form minimum mass for hub impacts on radial nets, and the material-utilization factor η as the single quantity through which topology enters;
3. the transmission of tension waves through a junction of N threads, including an exact finite-amplitude solution with shocks, rarefaction fans, and slack threads, which shows that the linear relations predict the failure threshold at the hub but underestimate the load on the neighbours, because same-side threads go slack, and that, in the longitudinal limit, localization at the hub is set by the number of threads rather than by stiffening; the transverse impedance of the off-axis threads, validated numerically, raises the load on the loaded thread by up to 15% for the compliant material; and the dynamic overload after a sudden thread break for power-law responses;
4. a dimensionless localization criterion and the mass saving $m_{\min}^A/m_{\min}^B$ due to sacrificial failure during the capture stroke, for a stiffening and a linear material, verified with explicit simulations, together with the mechanism that suppresses the saving for the linear material and the growth of the minimum mass towards the frame;

5. a test of spectral descriptors as predictors of η , and a graph-neural-network surrogate that predicts thread failure on unseen topologies.

The paper is organized as follows. Section II presents the model. Section III treats the single thread and the impact regime, Sec. IV the quasi-static capacity of radial nets, and Sec. V sacrificial failure. Section VI describes the simulator and Sec. VII the learned surrogate. Results are presented in Sec. VIII and discussed in Sec. IX; Sec. X concludes.

II. PHYSICAL MODEL

A. Net, projectile, and arrest criteria

The net is a planar graph $G = (\mathcal{V}, \mathcal{E})$ with a set $\mathcal{A} \subset \mathcal{V}$ of anchored nodes on a rigid frame. Each edge e is a thread of rest length ℓ_e , cross-section A_e , and density ρ , so that the linear mass density is $\mu_{0,e} = \rho A_e$ and the net mass is

$$m_{\text{net}} = \sum_{e \in \mathcal{E}} \rho A_e \ell_e. \quad (1)$$

The net is prestressed to a uniform strain ϵ_p (Sec. IID). The drone is a rigid sphere of mass M and radius r_d , hitting the net perpendicularly with speed v_0 at a point $\mathbf{p}$ of an admissible region $\mathcal{P}$ (e.g. the central disk of radius $R/2$). We compare two arrest criteria:

- **A (no failure):** the drone is stopped and no thread breaks, $\epsilon_e(t) < \epsilon_b$ for all e and t ;
- **B_{any} (sacrificial failure, single-use net):** the drone is stopped and does not perforate the net; threads may break, with fewer than k_{max} failed threads (a thread counts once, however many of its segments fail);
- **B_{loc} (localized sacrificial failure, repairable net):** criterion B_{any} holds and the first failure of every failed thread lies within a distance R_{max} of the in-plane position of the drone at that time (Sec. V C).

Criterion B_{any} asks only that the net does its job once. Criterion B_{loc} also asks that the damage stays near the impact, so that the net can be repaired or survive a second hit. The distance is measured from the drone and not from the initial impact point because the drone drifts in the plane of the net during the arrest (Sec. IV B). A net that satisfies A satisfies both B criteria, so $m_{\text{min}}^{\text{B}_{\text{loc}}} \leq m_{\text{min}}^{\text{A}}$ and $m_{\text{min}}^{\text{B}_{\text{any}}} \leq m_{\text{min}}^{\text{B}_{\text{loc}}}$. The difference between the two B criteria is the cost of requiring localized damage. The minimum mass under criterion $X \in \{A, B_{\text{any}}, B_{\text{loc}}\}$ is $m_{\text{min}}^X = \min m_{\text{net}}$ such that criterion X holds for every $\mathbf{p} \in \mathcal{P}$ (worst case).

Capture stroke. All criteria are evaluated over the *capture stroke*: the drone is stopped when its velocity normal to the net first reverses, and the conditions above apply up to that instant (Algorithm 1). The model has no dissipation other than thread failure and the capture of the gripped node, so after the reversal the net returns its elastic energy to the drone and the system keeps oscillating. Whether a net that captures the drone also *retains* it is therefore a question about dissipation (material hysteresis, aerodynamic drag, rotor thrust), which is outside the present model; Sec. VIII reports how often a further thread breaks after the reversal.

Contact model. We assume that the thread hit by the drone becomes entangled at the contact point (in the rotors, arms, or landing gear) and does not slide over the drone afterwards. In the model, the net node closest to $\mathbf{p}$ is attached to the drone at first contact; every other net node interacts with the drone through a frictionless penalty force (Sec. VI). With this assumption a thread hit at a distance a from the hub is split into two segments that carry different tensions and can fail independently (Fig. 1c). The opposite limit, frictionless sliding of the thread over the drone, equalizes the tension on both sides of the contact; it is computed as a sensitivity case in Sec. VIII. The drone starts tangent to the net, gravity is neglected (the net is vertical and the drone flies horizontally into it, so its weight acts in the plane of the net; for a horizontal net the work Mgw_{max} would add up to $\sim 15\%$ to the energy to absorb, ?), and the net is resolved finely enough that the spacing of the discretization nodes satisfies $\ell_e/n_s \leq r_d/4$.

B. Constitutive law and model materials

We model the thread with the simplest law that captures strain stiffening, a linear term plus a cubic correction; the nominal tension is

$$T(\epsilon) = E_0 A \epsilon + \beta \epsilon^3, \quad \epsilon < \epsilon_b, \quad (2)$$

TABLE I. Model materials. For S, b is fixed by $\sigma(\epsilon_b) = \sigma_b$. Speeds: $c_{L,0} = \sqrt{E_0/\rho}$ (small-strain longitudinal), $c_L(\epsilon_b)$ from Eq. (7), $c_T(\epsilon_b)$ from Eq. (8); v_c from Eq. (10).

Quantity	S (silk-like)	D (Dyneema-like)	Source / comment
Density ρ (kg/m ³)	1300	975	typical; [TO DO: check]
Initial modulus E_0 (GPa)	2	110	D: [14]; S: model choice
Cubic coefficient b (GPa)	22.2	0	S: from σ_b
Failure strain ϵ_b	0.30	0.032	typical
Failure stress σ_b (GPa)	1.20	3.52	
Tangent modulus at ϵ_b (GPa)	8.0	110	stiffening ratio 4 for S
e_{mat} (kJ/kg)	103.8	57.8	Eq. (3)
$c_{L,0}$ (m/s)	1240	10 622	
$c_L(\epsilon_b)$ (m/s)	1754	10 622	
$c_T(\epsilon_b)$ (m/s)	843	1870	
Critical velocity v_c (m/s)	936	1094	instant failure at impact

and the thread breaks when $\epsilon = \epsilon_b$. We write $\sigma(\epsilon) = T/A = E_0\epsilon + b\epsilon^3$ with $b = \beta/A$, and define the strain-energy function and the specific energy to failure

$$\Phi(\epsilon) = \int_0^\epsilon \sigma d\epsilon' = \frac{E_0\epsilon^2}{2} + \frac{b\epsilon^4}{4}, \quad e_{\text{mat}} = \frac{\Phi(\epsilon_b)}{\rho}. \quad (3)$$

Two model materials are used (Table I): a silk-like stiffening material **S** and a Dyneema-like linear-brittle material **D** ($b = 0$). They are *model* materials: the parameters are chosen to be representative, not fitted to a specific fiber.

Tavi's note: Rareş and Antonia should also fit E_0 and b to the silk law used by Cranford *et al.* (Supplementary Information of [1]) and to the Dyneema data of [14], and check that the conclusions do not change. The cubic law has no softening (yield) region, unlike real dragline silk; this is a stated limitation.

C. Drone parameters

We consider $M = 0.25\text{--}2$ kg and $v_0 = 10\text{--}20$ m/s, i.e. kinetic energies $E_{\text{kin}} = \frac{1}{2}Mv_0^2 = 12.5\text{--}400$ J. The drone is much larger than the mesh size: we require the largest free opening of the net to be smaller than r_d , which sets a minimum number of threads for a given frame radius R but, as shown in Sec. IV, not a minimum mass.

D. Prestress and the force-density method

For a net of perfectly flexible threads in equilibrium, define the force density $q_e = T_e/l_e$ (tension over current length). Nodal equilibrium $\sum_{e \ni i} q_e(\mathbf{x}_j - \mathbf{x}_i) = \mathbf{0}$ at every free node i is linear in the coordinates and can be written as

$$L_q^{ff} \mathbf{x}_f = -L_q^{fa} \mathbf{x}_a, \quad (4)$$

where L_q is the graph Laplacian weighted by q_e and the superscripts f, a denote free and anchored nodes [15, 16]. Given q_e and the anchor positions, Eq. (4) gives the prestressed geometry; the rest lengths then follow from $\ell_e = l_e/(1 + \epsilon_e)$ with $T(\epsilon_e) = q_e l_e$. Small out-of-plane vibrations $u_i(t)$ about this state obey

$$L_q^{ff} \phi = \omega^2 \mathbf{M} \phi, \quad (5)$$

with $\mathbf{M}$ the lumped nodal mass matrix. Equation (5) is the link to the spectral part of the work (Sec. VII).

Tavi's note: For the star net with uniform prestress, Eq. (4) gives the hub at the centre, as expected. Rareş and Antonia should derive Eq. (5) from Newton's law for the transverse displacement of a node connected to its neighbours by strings of tension T_e (the restoring force is $\sum_e T_e(u_j - u_i)/l_e$), and verify it for N radials with a single hub mass: $\omega^2 = NT_p/(R m_{\text{hub}})$.

Reference solution (Tavi only). Let node i have transverse (out-of-plane) displacement u_i and lumped mass m_i , and let edge $e = (i, j)$ carry tension T_e over current length l_e . To first order in u the tensions and in-plane positions do not change (the length change is $O(u^2)$), and the transverse component of the force exerted by edge e on node i is $T_e(u_j - u_i)/l_e = q_e(u_j - u_i)$. Newton's law gives $m_i \ddot{u}_i = \sum_{e \ni i} q_e(u_j - u_i)$, i.e. $M \ddot{\mathbf{u}}_f = -L_q^{ff} \mathbf{u}_f$ with $u = 0$ at anchored nodes. With $\mathbf{u}_f = \phi e^{i\omega t}$ this is Eq. (5). For the star with massless radials and hub mass m_{hub} : one free node, $L_q^{ff} = \sum_{k=1}^N q_k = NT_p/R$, hence $\omega^2 = NT_p/(Rm_{\text{hub}})$. Note that the in-plane problem has a different stiffness (it involves $T'(\epsilon)$, not T/l); Eq. (5) is the out-of-plane spectrum only.

E. Dimensionless groups

To characterize the dynamic response of the net during drone arrest, we identify five key dimensionless parameters that govern the physics of the impact:

$$\Pi_E = \frac{v_0^2}{2e_{\text{mat}}}, \quad \Pi_T = \frac{v_0}{c_T(\epsilon_b)}, \quad \bar{\epsilon} = \frac{\epsilon_p}{\epsilon_b}, \quad S = \frac{b\epsilon_b^2}{E_0}, \quad \bar{r} = \frac{r_d}{R}. \quad (6)$$

Each parameter isolates a specific physical aspect of the system:

- **Energy capacity ratio (Π_E):** Measures the kinetic energy of the drone relative to the maximum failure energy density of the material. Ideally ($\eta = 1$), Π_E represents the theoretical minimum mass ratio m_{net}/M required to stop the drone.
- **Transverse wave parameter (Π_T):** Compares the impact velocity v_0 with the fastest transverse wave speed $c_T(\epsilon_b)$ at failure. A ratio $\Pi_T \ll 1$ indicates that transverse stress waves travel much faster than the drone penetration, placing the impact deep within a quasi-static arrest regime.
- **Prestrain ratio ($\bar{\epsilon}$):** Normalizes the initial prestrain ϵ_p against the breaking strain ϵ_b , governing the initial geometric stiffness and tension in the net.
- **Stiffening parameter (S):** Quantifies the non-linearity of the constitutive law, measuring the ratio between non-linear cubic stiffening and initial linear elasticity ($S = 0$ for the linear material D, $S = 0.99$ for the silk-like material S).
- **Geometric ratio ($\bar{r}$):** Compares the drone radius r_d with the frame radius R , defining the spatial scale of the contact zone relative to the net size.

In the quasi-static regime analyzed in this work, the structural mass of the net is much smaller than the drone mass ($m_{\text{net}} \ll M$). Consequently, the inertia ratio $M/(\rho R^3)$ decouples from the energy absorption mechanics, and the dimensionless set in Eq. (6), together with the breaking strain ϵ_b and the network topology, fully characterizes the arrest process.

Reference solution (Tavi only). *Variables.* In the arrest problem the output is m_{min} and the inputs are $M, v_0, R, r_d, \rho, E_0, b$, and the dimensionless $\epsilon_b, \epsilon_p, N$ (plus the topology). The eight dimensional quantities $m_{\text{min}}, M, v_0, R, r_d, \rho, E_0, b$ involve three base dimensions (M, L, T), so there are $8 - 3 = 5$ independent groups, for example

$$\frac{m_{\text{min}}}{M}, \quad \frac{\rho v_0^2}{E_0}, \quad \frac{b}{E_0}, \quad \frac{r_d}{R}, \quad \frac{M}{\rho R^3},$$

together with $\epsilon_b, \epsilon_p, N$. *Relation with Eq. (6).* With $s = b\epsilon_b^2/E_0$, $\Pi_E = \rho v_0^2/[E_0\epsilon_b^2(1 + s/2)]$ and $\Pi_T^2 = \rho v_0^2(1 + \epsilon_b)/[E_0\epsilon_b(1 + s)]$, so $\Pi_T^2/\Pi_E = \epsilon_b(1 + \epsilon_b)(1 + s/2)/(1 + s)$: the pair (Π_E, Π_T) at fixed s is equivalent to $(\rho v_0^2/E_0, \epsilon_b)$. Equation (6) therefore contains $\rho v_0^2/E_0, \epsilon_b$ (implicitly), $b/E_0, \epsilon_p/\epsilon_b, r_d/R$, but *not* $M/(\rho R^3)$ and *not* N . *Conclusion.* The set is complete only in the quasi-static, massless-thread limit, where $M/(\rho R^3)$ drops out because the stored energy and m_{net} are both proportional to the thread volume (this is why m_{min} does not depend on R , Sec. IV). For the full dynamic problem $M/(\rho R^3)$ must be added; N and the topology enter separately; gravity would add gR/v_0^2 (neglected). Suggest writing ϵ_b explicitly in Eq. (6).

Table II. $E_{\text{kin}} = \frac{1}{2}Mv_0^2$; $\Pi_E = v_0^2/(2e_{\text{mat}})$ with e_{mat} from Table I; ϵ_w is the root of Eq. (9), well approximated by $\epsilon_w \simeq (\rho v_0^2/2E_0)^{2/3}$ since $\epsilon_w \ll 1$; $c_T(\epsilon_b) = \sqrt{\sigma_b/[\rho(1 + \epsilon_b)]}$ (843 and 1870 m/s); $m_{\text{min}}^{\eta=1} = E_{\text{kin}}/e_{\text{mat}}$. Check: S, $M = 1$ kg, $v_0 = 15$ m/s: $\Pi_E = 225/(2 \times 103\,800) = 1.08 \times 10^{-3}$, $\epsilon_w \simeq (1300 \times 225/4 \times 10^9)^{2/3} = 1.75 \times 10^{-3} = 0.58\% \epsilon_b$ (exact root 0.59%), $c_T/v_0 = 56$, $m = 112.5/103.8 = 1.08$ g.

III. SINGLE THREAD AND IMPACT REGIME

A. Transverse impact on a thread with the cubic law

Consider a long thread along x , at rest and unstrained, whose point $x = 0$ is suddenly given a constant transverse velocity v_0 . We use the Lagrangian coordinate X and the notation of Smith *et al.* [3]. For a stiffening law the longitudinal front is a shock; mass and momentum conservation across it give

$$u_2 = c_L \epsilon, \quad T(\epsilon) = \mu_0 c_L u_2 \quad \implies \quad c_L = \sqrt{\frac{T(\epsilon)}{\mu_0 \epsilon}}, \quad (7)$$

where u_2 is the particle velocity towards the impact point behind the front. Behind the longitudinal front the strain is uniform and equal to ϵ ; it is not changed by the transverse kink, which travels in material coordinates at

$$c_T = \sqrt{\frac{T(\epsilon)}{\mu_0(1+\epsilon)}}. \quad (8)$$

B. Kinematic relation

At time t the longitudinal front is at $X = c_L t$ and the kink at $X = c_T t$. A material point of region 2 ($c_T t < X < c_L t$) has been displaced towards the origin by $\epsilon(c_L t - X)$, so the kink is at the lab position $x_K = (1+\epsilon)c_T t - \epsilon c_L t$. Region 1 is a straight segment of stretched length $(1+\epsilon)c_T t$ from $(0, v_0 t)$ to $(x_K, 0)$. Pythagoras' theorem then gives

$$v_0^2 = (1+\epsilon)^2 c_T^2 - [(1+\epsilon)c_T - \epsilon c_L]^2 = \frac{T(\epsilon)}{\mu_0} [2\sqrt{\epsilon(1+\epsilon)} - \epsilon]. \quad (9)$$

For a linear law, $T/\mu_0 = c^2 \epsilon$ and Eq. (9) reduces to Smith's classical result $v_0 = c\sqrt{2\epsilon\sqrt{\epsilon(1+\epsilon)} - \epsilon^2}$ [3]. For $\epsilon \gg 1$ with $T \simeq \beta \epsilon^3$, Eq. (9) gives $v_0^2 \simeq \beta \epsilon^4 / \mu_0$, so $\epsilon \propto v_0^{1/2}$ and $T_1 \propto v_0^{3/2}$ in the strongly stiffening limit.

Tavi's note: Check this point carefully with Rareş and Antonia. The reference solution (Part B.2) of the problem that motivated this study places the kink at $x_K = (1+\epsilon_2)c_T t$, i.e. it omits the displacement $-\epsilon c_L t$ of region 2 towards the impact point, and it allows $\epsilon_1 \neq \epsilon_2$. In Smith's theory the strain is continuous across the kink ($\epsilon_1 = \epsilon_2$). Both versions give the same asymptotic scaling $T_1 \propto v_0^{3/2}$, so the scaling result (B.3) is unaffected, but the full relation (9) differs. Rareş and Antonia must redo the derivation independently and decide which is correct; the simulator (test 1, Sec. VIII A) settles the question.

C. Critical velocity

The thread breaks immediately at impact when the strain given by Eq. (9) reaches ϵ_b :

$$v_c = \sqrt{\frac{\sigma_b}{\rho} [2\sqrt{\epsilon_b(1+\epsilon_b)} - \epsilon_b]}, \quad (10)$$

which gives $v_c \approx 936$ m/s for S and ≈ 1094 m/s for D (Table I).

D. Impact regime of drones

Table II evaluates Eqs. (6) and (9) for drones. Three conclusions follow.

1. The strain carried by the initial waves, $\epsilon_w(v_0)$, is 0.2–0.9% of the failure strain. Drones cannot break a thread by wave loading alone, in contrast to ballistic impacts ($v_0 \gtrsim v_c$).
2. $c_T(\epsilon_b)/v_0 \approx 40$ –190: during arrest, transverse waves cross the net tens of times. The response is quasi-static in the sense that the net shape follows the projectile, and the strain field is set by geometry rather than by wave fronts.

TABLE II. Impact regime for drones, for materials S and D. ϵ_w : strain behind the initial waves from Eq. (9); $m_{\min}^{\eta=1} = E_{\text{kin}}/e_{\text{mat}}$ is the absolute lower bound of the net mass. Analytical values computed by Tavi; Rareş and Antonia recompute them.

M (kg)	v_0 (m/s)	E_{kin} (J)	S (silk-like)				D (Dyneema-like)			
			Π_E (10^{-3})	ϵ_w/ϵ_b (%)	c_T/v_0	$m_{\min}^{\eta=1}$ (g)	Π_E (10^{-3})	ϵ_w/ϵ_b (%)	c_T/v_0	$m_{\min}^{\eta=1}$ (g)
0.25	10	12.5	0.48	0.34	84	0.12	0.87	0.18	187	0.22
0.25	20	50	1.93	0.87	42	0.48	3.46	0.46	94	0.87
1	15	112.5	1.08	0.59	56	1.08	1.95	0.31	125	1.95
2	10	100	0.48	0.34	84	0.96	0.87	0.18	187	1.73
2	20	400	1.93	0.87	42	3.85	3.46	0.46	94	6.92

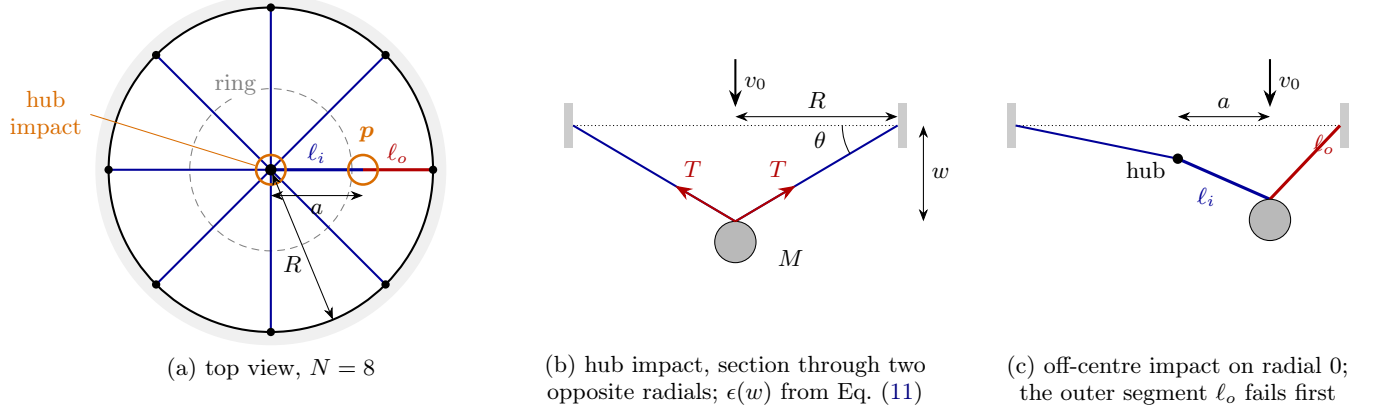


FIG. 1. Star net. (a) Top view: N radials of rest length ℓ anchored on a rigid frame of radius R and joined at the hub; the dashed circle is the optional ring at $R/2$ (Sec. IV B). Circles mark the two impact configurations. For an off-centre impact at distance a , radial 0 is split into an inner segment ℓ_i and an outer, anchored segment ℓ_o (red). (b) Hub impact: all radials carry the same tension $T(\epsilon(w))$ at inclination $\theta = \arctan(w/R)$, and the restoring force is Eq. (12). (c) Off-centre impact: the hub is held by the other $N - 1$ radials, the short outer segment is strained most and is the sacrificial element. Deflections are schematic; for material S the deflection at failure is large, $w_b/R \approx 0.65\text{--}0.83$ (Table III).

- $\Pi_E \sim 10^{-3}$: even at perfect utilization the net is two to three orders of magnitude lighter than the drone (0.1–7 g of thread for 12.5–400 J), so thread inertia is negligible.

The wave picture of Smith therefore controls only two local, fast events: the first microseconds after contact and the release wave emitted when a thread breaks (Sec. V).

Tavi’s note: This regime analysis changes the framing of the earlier planning documents, where η was assumed to be limited by wave propagation. For drones this is not the case: topology enters through the quasi-static strain distribution. Wave effects matter again only at thread failure. The research question “is there a speed regime in which topology no longer matters?” is answered here in reverse: the wave-limited regime starts near $v_0 \sim \sqrt{e_{\text{mat}}} \sim 250\text{--}320$ m/s, far above drone speeds.

IV. QUASI-STATIC CAPACITY OF RADIAL NETS

A. Hub impact on a star net

Consider N radials of equal cross-section, anchored on a circle of radius R and prestressed to ϵ_p , and a drone hitting the hub (Fig. 1a,b). At transverse deflection w every radial has the same strain,

$$1 + \epsilon(w) = (1 + \epsilon_p) \sqrt{1 + (w/R)^2} \simeq 1 + \epsilon_p + \frac{w^2}{2R^2}, \quad (11)$$

and the restoring force and stored energy are

$$F(w) = N T(\epsilon(w)) \frac{w}{\sqrt{R^2 + w^2}}, \quad U(w) = N \ell A [\Phi(\epsilon(w)) - \Phi(\epsilon_p)], \quad (12)$$

TABLE III. Hub impact on a star net: utilization η_A from Eq. (13) and stopping distance w_b/R at failure, analytical and numerical.

Material	ϵ_p/ϵ_b	η_A analytical	η_A numerical	w_b/R analytical	w_b/R numerical
S	0	1.000	0.999	0.831	0.831
S	0.1	0.993	0.992	0.770	0.770
S	0.3	0.937	0.936	0.650	0.650
D	0	1.000	0.995	0.255	0.255
D	0.1	0.990	0.988	0.241	0.241
D	0.3	0.910	0.910	0.212	0.212

with $\ell = R/(1 + \epsilon_p)$ the rest length. Since the net mass is $m_{\text{net}} = N\rho A\ell$, the energy absorbed up to failure per unit net mass is $e_{\text{mat}} \eta_A$ with

$$\boxed{\eta_A = 1 - \frac{\Phi(\epsilon_p)}{\Phi(\epsilon_b)}}, \quad m_{\text{min}}^A = \frac{E_{\text{kin}}}{\eta_A e_{\text{mat}}}. \quad (13)$$

The minimum mass does not depend on N or on R . The number of radials is set only by the mesh (the drone must not pass through), and R only by the allowed stopping distance, $w_b/R = \sqrt{[(1 + \epsilon_b)/(1 + \epsilon_p)]^2 - 1}$ (Table III).

Tavi's note: Result (13) is exact within the quasi-static, massless-thread model and is a strong baseline: a hub impact on a star net uses the material perfectly. For S, prestress costs almost nothing ($\eta_A = 0.993$ at $\epsilon_p = 0.1\epsilon_b$) because Φ is small at small strain. The whole topology question therefore lives in the *off-centre* impacts of Sec. IV B, where the strain is non-uniform. Note that for S the stopping distance is large ($w_b \approx 0.8R$), so the small-angle form of Eq. (11) must not be used for S.

The derivation of Eqs. (11)–(13) and its assumptions are given in Appendix B. **[TO DO: Rareş and Antonia check the derivation in Appendix B, plot $F(w)$ for S and D (Fig. 4), and compute the arrest deflection w_{max} from $U(w_{\text{max}}) = E_{\text{kin}}$ for the cases of Table II.]**

Reference solution (Tavi only). The derivation and its checks are in Appendix B; the analytical $F(w)$ of Eq. (12) agrees with the quasi-static simulation in Fig. 4, and the failure deflections w_b/R to three decimals (Table III). The arrest deflection is the closed form of the reference solution at the end of Appendix B. It depends only on m_{net}/m^* with $m^* = E_{\text{kin}}/e_{\text{mat}}$, so it is the same for all five rows of Table II once the net mass is chosen in proportion to E_{kin} : at $m_{\text{net}} = m_{\text{min}}^A$, $w_{\text{max}} = w_b$, i.e. $w_{\text{max}}/R = 0.770$ for S and 0.241 for D at $\epsilon_p = 0.1\epsilon_b$. The TODO is therefore answered by the table $w_{\text{max}}(\mu)$ of that reference solution, with $\mu = m_{\text{net}}/m_{\text{min}}^A$.

B. Off-centre impact on a radial

Let the drone hit radial 0 at distance a from the hub (Fig. 1a,c). Radial 0 is split into an outer segment of rest length $\ell_o \propto R - a$, anchored on the frame, and an inner segment $\ell_i \propto a$ attached to the hub, which is in turn held by the other $N - 1$ radials. The outer segment is shorter and anchored, so for a given deflection it is strained more: it fails first. Under criterion A this failure ends the capacity of the net, and the utilization drops to

$$\eta_A(a) = \frac{U_{\text{net}}(w_b(a))}{m_{\text{net}} e_{\text{mat}}} < 1, \quad (14)$$

because most threads are far below ϵ_b when the outer segment fails. **[TO DO: Rareş and Antonia (i) write the quasi-static equilibrium for this configuration, treating the hub and the other radials as a nonlinear spring along radial 0; (ii) obtain $\eta_A(a)$ for S and D, first for $\epsilon_p = 0$; (iii) add one ring (spiral) at radius $R/2$ and repeat numerically. Results go in Fig. 5 and Table XIV.]**

Reference solution (Tavi only). *Model.* Radial 0 lies along x ; the drone is entangled with the net at $\mathbf{p} = (a, 0, w)$ and moves along z only (lateral drift neglected, $M \gg m_{\text{net}}$). The hub $\mathbf{h} = (x_h, 0, z_h)$ is free (by symmetry $y_h = 0$). Segments: outer, from $\mathbf{p}$ to the anchor $(R, 0, 0)$, rest length $(R - a)/(1 + \epsilon_p)$; inner, from $\mathbf{p}$ to $\mathbf{h}$, rest length $a/(1 + \epsilon_p)$;

radials $k = 1, \dots, N - 1$, from $\mathbf{h}$ to $R(\cos \varphi_k, \sin \varphi_k, 0)$, rest length $R/(1 + \epsilon_p)$. Threads carry no compression ($\Phi = 0$ for $\epsilon < 0$). At each w the quasi-static hub position minimizes

$$\mathcal{U}(\mathbf{h}; w) = A \sum_e \ell_e \Phi(\epsilon_e(\mathbf{h}, w)),$$

which is the “nonlinear spring” of the TODO: the other radials and the hub act on the inner segment through $\partial \mathcal{U} / \partial \mathbf{h} = 0$. The outer strain $\epsilon_o = (1 + \epsilon_p) \sqrt{1 + w^2 / (R - a)^2} - 1$ does not depend on $\mathbf{h}$; the inner strain does. The first segment to reach ϵ_b fails at w_1 , and $\eta_A(a) = \mathcal{U}(w_1) / (m_{\text{net}} e_{\text{mat}})$ with $m_{\text{net}} = \rho A N R / (1 + \epsilon_p)$. Under criterion B the failed segment is removed, the hub re-equilibrates, and loading continues to the next failure at w_2 ; the energy absorbed from the drone is $W_B = \mathcal{U}(w_1) + \mathcal{U}_{\text{rem}}(w_2) - \mathcal{U}_{\text{rem}}(w_1)$ and $\eta_B = W_B / (m_{\text{net}} e_{\text{mat}})$. There is no closed form; the values below come from a direct minimization (Nelder–Mead, step $\Delta w = 0.002R$ for S and $0.0005R$ for D, failure located by bisection; script `ref/offc.py`).

a/R	S			D		
	first failure	η_A	η_B	first failure	η_A	η_B
0.05	inner	0.176	0.261	inner	0.224	0.299
0.10	inner	0.198	0.257	inner	0.217	0.270
0.20	outer	0.206	0.251	outer	0.218	0.248
0.30	outer	0.128	0.248	outer	0.135	0.239
0.40	outer	0.090	0.246	outer	0.093	0.234
0.50	outer	0.068	0.244	outer	0.069	0.231

$N = 8$, $\epsilon_p = 0$. For $\epsilon_p = 0.1\epsilon_b$ the values increase by 3–16%, e.g. $\eta_A(0.5R) = 0.075$ (S), 0.078 (D); here and below the prestress energy is subtracted from the absorbed energy, as in Eq. (13).

Dependence on N ($\epsilon_p = 0$, $a = 0.5R$): $\eta_A = 0.132, 0.068, 0.035$ for $N = 4, 8, 16$ (S), and practically the same for D; $\eta_A \propto 1/N$, because the other radials are barely strained when the loaded radial fails (max radial strain $0.11\epsilon_b$ at $a = 0.5R$, $N = 8$). *Lateral freedom of the drone.* The computation above holds the gripped point at $(a, 0)$, which requires a lateral force on the drone. A real drone receives no lateral force except from the net, so it can drift in the plane. Adding the lateral position p_x of the gripped point as an equilibrium unknown (zero lateral force) gives the opposite limit:

a/R	S			D		
	η_A fixed	η_A free	p_x/R	η_A fixed	η_A free	p_x/R
0.10	0.198	0.200	0.098	0.217	0.268	0.097
0.20	0.206	0.230	0.237	0.218	0.236	0.202
0.30	0.128	0.208	0.382	0.135	0.215	0.307
0.40	0.090	0.168	0.519	0.093	0.194	0.412
0.50	0.068	0.136	0.649	0.069	0.180	0.517

$N = 8$, $\epsilon_p = 0$, p_x at first failure. The simulator reproduces both columns within 0.5%.

The gripped point slides outward, toward the anchor of the loaded radial, until the tensions of the inner and outer segments balance. This doubles η_A at $a = R/2$. The dynamic impact lies between the two limits, because the drone inertia limits the drift during the arrest: in the reference case (S, $M = 1$ kg, $v_0 = 15$ m/s, $a = R/2$) the drone reaches $x = 0.61R$ at arrest, and the dynamic m_{min}^A at $\epsilon_p = 0$ corresponds to $\eta_A \approx 0.105$. At fixed E_{kin} the drift is nearly independent of M : the lateral force scales with the net tension, the arrest time with $\sqrt{M/E_{\text{kin}}}$, and the drift with their product divided by M . *Conclusions for the paper.* (i) Off-centre impacts on a star are very inefficient: at $a = R/2$ only 7–14% of the material is used (fixed and free lateral position of the drone), against 100% for the hub impact. (ii) With the gripped point fixed, S and D give almost the same $\eta_A(a)$: the utilization is set by geometry, not by the constitutive law. With the point free, D is 30% better than S at $a = R/2$. (iii) The ring of step (iii) is needed to redistribute load; its effect has to come from the simulator. (iv) The results assume the thread is gripped at $\mathbf{p}$ (entangled in the rotors). If it slides freely over the drone, the tension is the same in the inner and outer segments, radial 0 fails as a whole, and in a pure star $\eta_B = \eta_A$: there is no sacrificial saving without a ring. The contact model must be fixed in Sec. II and in the simulator.

Tavi’s note: This is the configuration of Cranford *et al.*, who load one radial thread. The expected qualitative picture is that the loaded outer segment is the sacrificial element. The quantitative question is how much energy the net can still absorb after it breaks (Sec. V).

TABLE IV. Linear transmission at a hub of N equally spaced threads: tension in the loaded thread at the hub, largest increase and largest decrease among the other threads, all normalized by T_{inc} . 1D: Eq. (15), longitudinal waves only. 2D: Eq. (16), with the transverse impedance of the off-axis threads. Numerical: Algorithm 1 with a single small incident pulse, material S, $\epsilon_p = 0.1 \epsilon_b$ ($Z_T/Z_L = 0.169$).

N	S_N	S'_N	T_0			$\max \Delta T_j$			$\min \Delta T_j$		
			1D	2D	num.	1D	2D	num.	1D	2D	num.
2	1	1.000	1.000	1.000	1.000	1.000	1.000	1.000	—	—	—
3	0.5	0.757	0.667	0.862	0.874	0.667	0.569	0.577	—	—	—
4	1	1.343	1.000	1.146	1.144	1.000	0.854	0.856	0.000	0.000	0.013
6	2	2.515	1.333	1.431	1.430	0.667	0.569	0.571	-0.333	-0.285	-0.281
8	3	3.686	1.500	1.573	1.574	0.500	0.427	0.429	-0.354	-0.302	-0.301
12	5	6.029	1.667	1.715	1.718	0.333	0.285	0.286	-0.289	-0.246	-0.247
16	7	8.372	1.750	1.787	1.789	0.250	0.213	0.214	-0.231	-0.197	-0.198

V. SACRIFICIAL FAILURE

A. Tension waves at a junction of N threads

When a thread breaks or is suddenly loaded, a tension wave reaches the next junction. Consider a massless junction (hub) of N equally spaced threads of equal impedance $Z = \mu_0 c_L$ (linear regime, small hub displacement), with an incident tension wave of amplitude T_{inc} in thread 0. Let $\varphi_j = 2\pi j/N$ be the angle between thread j and thread 0, and V the hub velocity along thread 0. The tension in thread 0 at the hub is $T_0 = 2T_{\text{inc}} - ZV$, thread j is lengthened at the rate $-V \cos \varphi_j$ so that $\Delta T_j = -ZV \cos \varphi_j$, and equilibrium of the massless hub along thread 0 gives $T_0 = ZV S_N$ with $S_N = \sum_{j=1}^{N-1} \cos^2 \varphi_j$. Hence

$$\frac{T_0}{T_{\text{inc}}} = \frac{2S_N}{1 + S_N}, \quad \frac{\Delta T_j}{T_{\text{inc}}} = -\frac{2 \cos \varphi_j}{1 + S_N}, \quad S_N = \begin{cases} 1, & N = 2 \\ N/2 - 1, & N \geq 3. \end{cases} \quad (15)$$

In this longitudinal (1D) approximation the wave passes straight through for $N = 2$ and $N = 4$ ($T_0 = T_{\text{inc}}$). For large N the hub acts like a clamp: the tension in the loaded thread approaches $2T_{\text{inc}}$, while each other thread receives at most $2T_{\text{inc}}/(1 + S_N) \sim 4T_{\text{inc}}/N$ (Table IV). Threads on the same side as thread 0 ($\cos \varphi_j > 0$) are *unloaded*; they stay taut only if $T_p > |\Delta T_j|$.

Transverse impedance. Equation (15) ignores that the off-axis threads also resist the hub motion transversely. A hub velocity V along thread 0 has a component $V \sin \varphi_j$ normal to thread j , which launches a transverse wave in that thread; for small slopes the resisting force is $T_p V \sin \varphi_j / c_{T,p} = Z_T V \sin \varphi_j$, with $Z_T = \mu_0 c_{T,p}$ and $c_{T,p}$ the transverse speed at the prestress. Transverse waves are slow, but their impedance is not negligible: $Z_T/Z_L = c_{T,p}/c_{L,p} \approx \sqrt{\epsilon_p}$ for a linear material. Including this term in the equilibrium of the hub leaves the form of Eq. (15) unchanged, with

$$S_N \rightarrow S'_N = S_N + \frac{Z_T}{Z_L} \sum_{j=1}^{N-1} \sin^2 \varphi_j = S_N + \frac{Z_T}{Z_L} \frac{N}{2} \quad (N \geq 3). \quad (16)$$

The correction vanishes for $N = 2$ and raises the load on the loaded thread for $N \geq 3$: for material S at $\epsilon_p = 0.1 \epsilon_b$ ($Z_T/Z_L = 0.169$), $T_0/T_{\text{inc}} = 1.146$ instead of 1 for $N = 4$, and the largest neighbour increase for $N = 8$ drops from 0.500 to 0.427. For D ($Z_T/Z_L = 0.057$) the changes are 3–5%. The simulation agrees with Eq. (16) to within 1.5% for all N (Table IV), and the deviation from Eq. (15) scales as $Z_T/Z_L \propto \sqrt{\epsilon_p}$ (for D at $\epsilon_p/\epsilon_b = 0.02, 0.1, 0.3$: $T_0/T_{\text{inc}} = 1.512, 1.527, 1.545$ numerically, against 1.512, 1.527, 1.545 from Eq. (16) and 1.500 from Eq. (15), $N = 8$). Since Z_T/Z_L is larger for the more compliant material, the transverse term makes the hub more clamp-like, and hence more localizing, for S than for D.

Reference solution (Tavi only). *Nonlinear junction.* For the stiffening material the threads meeting at a hub are generally at different strains: in an off-centre impact the loaded thread has been stretched to ϵ_0 , while its neighbours are still close to the prestrain ϵ_p . The appropriate impedance is then not the one of Eq. (7), which is the chord speed from the *unstrained* state, but the impedance for a change of state on top of the current strain. For a jump from

TABLE V. Nonlinear junction for material S, $\epsilon_p = 0.1\epsilon_b$, Eqs. (18)–(20): impedance ratio $r = Z(\epsilon_0)/Z(\epsilon_p)$, tension in the loaded thread at the hub T_0/T_{inc} , largest tension increase in a neighbour $\max \Delta T_j/T_{\text{inc}} = 2/(r + S_N)$ (thread opposite to thread 0), and transmitted energy fraction $\mathcal{T}$. For material D, $r = 1$ and the values are those of the first row. Simulation (test 3 with thread 0 pre-stretched to ϵ_0): T_0/T_{inc} agrees with Eq. (19) with the transverse impedance of Eq. (16) added within 0.4% for $\epsilon_0 \leq 0.67\epsilon_b$ ($N = 4, 8, 16$) and within 0.15% at $\epsilon_0 = 0.83$ and $0.98\epsilon_b$ ($N = 16$, measurement window halved to exclude the arrival of the transverse front); the transmitted fraction $\mathcal{T}$ is not measured reliably and is not compared.

ϵ_0/ϵ_b	r	$N = 4$ ($S_N = 1$)			$N = 8$ ($S_N = 3$)			$N = 16$ ($S_N = 7$)		
		T_0	$\max \Delta T_j$	$\mathcal{T}$	T_0	$\max \Delta T_j$	$\mathcal{T}$	T_0	$\max \Delta T_j$	$\mathcal{T}$
0.10	1.000	1.000	1.000	1.000	1.500	0.500	0.750	1.750	0.250	0.438
0.33	1.138	0.936	0.936	0.996	1.450	0.483	0.797	1.720	0.246	0.481
0.50	1.303	0.868	0.868	0.983	1.394	0.465	0.845	1.686	0.241	0.529
0.67	1.505	0.798	0.798	0.959	1.332	0.444	0.890	1.646	0.235	0.582
0.83	1.730	0.733	0.733	0.929	1.269	0.423	0.928	1.604	0.229	0.636
1.00	1.970	0.673	0.673	0.893	1.207	0.402	0.957	1.561	0.223	0.686

strain ϵ_1 to ϵ_2 , mass and momentum balance across the front give, as in Eq. (7),

$$T(\epsilon_2) - T(\epsilon_1) = Z_{12} \Delta u, \quad Z_{12} = \mu_0 c_{12}, \quad c_{12}^2 = \frac{T(\epsilon_2) - T(\epsilon_1)}{\mu_0(\epsilon_2 - \epsilon_1)}, \quad (17)$$

with Δu the jump in particle velocity. For a small increment on top of ϵ this reduces to the tangent impedance

$$Z(\epsilon) = \mu_0 c_t(\epsilon), \quad c_t^2(\epsilon) = \frac{T'(\epsilon)}{\mu_0} = \frac{E_0 + 3b\epsilon^2}{\rho}, \quad (18)$$

so that $\delta T = Z \delta u = E_t A \delta \epsilon$ with the tangent modulus $E_t = E_0 + 3b\epsilon^2$. Equation (7) is the special case $\epsilon_1 = 0$. For S the difference matters: at ϵ_b the tangent speed is $c_t = 2480$ m/s, whereas the chord speed from zero strain in Table I is 1754 m/s.

Let an incident tension increment T_{inc} arrive at the hub along thread 0, whose impedance is $Z_0 = Z(\epsilon_0)$, while the other $N - 1$ threads have impedance $Z_j = Z(\epsilon_p)$. The derivation leading to Eq. (15) carries over with two changes: the tension in thread 0 at the hub is $T_0 = 2T_{\text{inc}} - Z_0 V$, and $\Delta T_j = -Z_j V \cos \varphi_j$. Hub equilibrium along thread 0 still gives $T_0 = Z_j V S_N$, so $V = 2T_{\text{inc}}/(Z_0 + Z_j S_N)$ and, with the impedance ratio $r = Z_0/Z_j$,

$$\frac{T_0}{T_{\text{inc}}} = \frac{2S_N}{r + S_N}, \quad \frac{\Delta T_j}{T_{\text{inc}}} = -\frac{2 \cos \varphi_j}{r + S_N}, \quad \frac{T_{\text{ref}}}{T_{\text{inc}}} = \frac{S_N - r}{S_N + r}, \quad (19)$$

where $T_{\text{ref}} = T_0 - T_{\text{inc}}$ is the reflected increment. Equation (15) is recovered for $r = 1$. The energy flux carried by a linear wave is $\delta T^2/Z$, so the fractions of the incident energy transmitted to the neighbours and reflected into thread 0 are

$$\mathcal{T} = \sum_{j=1}^{N-1} \frac{\Delta T_j^2/Z_j}{T_{\text{inc}}^2/Z_0} = \frac{4rS_N}{(r + S_N)^2}, \quad \mathcal{R} = \left(\frac{S_N - r}{S_N + r} \right)^2, \quad \mathcal{T} + \mathcal{R} = 1. \quad (20)$$

The hub is impedance-matched for $r = S_N$: the neighbours act together as a single thread of impedance $S_N Z_j$.

For S, $r = \sqrt{E_t(\epsilon_0)/E_t(\epsilon_p)}$ grows from 1 to 1.97 as the loaded thread goes from ϵ_p to ϵ_b (Table V); for D, $r = 1$ at all strains. Three consequences follow. (i) *At a given incident tension*, stiffening of the loaded thread reduces the tension sent to each neighbour by the factor $(1 + S_N)/(r + S_N)$: by up to 33% for $N = 4$, 20% for $N = 8$, and 11% for $N = 16$. This is the localizing effect: the neighbours are pulled less. (ii) The partition of the hub reaction among the neighbours, $\Delta T_j/T_0 = -\cos \varphi_j/S_N$, is set by geometry alone and does not depend on r . Stiffening changes how much of the incident wave is reflected, not how the transmitted part is shared. (iii) *At a given incident strain* the conclusion is reversed: since $T_{\text{inc}} = E_t(\epsilon_0) A \delta \epsilon_{\text{inc}}$ and the neighbours respond with $E_t(\epsilon_p)$, the strain increment in neighbour j is $\delta \epsilon_j/\delta \epsilon_{\text{inc}} = -2r^2 \cos \varphi_j/(r + S_N)$, which grows with r . For $N \geq 8$ ($S_N \geq 3 > r$), stiffening also brings the hub closer to impedance matching and increases the energy fraction $\mathcal{T}$ transmitted to the neighbours, from 0.75 to 0.96 for $N = 8$. Because all threads share the same failure tension $T(\epsilon_b)$, the tension statement (i) is the relevant one for failure. The effect is moderate, however: the main reason why the neighbours survive is that they start from $T(\epsilon_p) \ll T(\epsilon_b)$, not the impedance mismatch.

Tavi's note: Assumptions of the nonlinear junction relations (reference solution), to be stated in the final text or checked: (a) the increments are small compared with the background, so that tangent impedances apply; for a finite incident jump use the chord impedance between the two states in thread 0 and in each loaded neighbour. (b) Neighbours with $\cos \varphi_j > 0$ are unloaded; for a stiffening law unloading waves are rarefaction fans, not shocks, and their impedance varies across the fan, but for small increments the tangent value is sufficient. They must also stay taut, $T(\epsilon_p) > |\Delta T_j|$. (c) The background state with $T(\epsilon_0) \neq T(\epsilon_p)$ is not an in-plane equilibrium of a flat star; it is maintained by the out-of-plane geometry and the drone contact, and we study only the in-plane increments. (d) Transverse waves and hub inertia are neglected, as for Eq. (15). The expected statement “the transmitted tension decreases as the loaded thread stiffens” holds in the tension sense (i), not in the strain or energy sense (iii).

[TO DO: Nonlinear junction. Rareş and Antonia generalize Eq. (15) to a strained incident thread and prestressed neighbours with strain-dependent impedance, obtain the transmitted and reflected tension and energy fractions, tabulate them for S, and discuss whether stiffening localizes the load. Then test 3 of Sec. VIII A is repeated with thread 0 pre-stretched to ϵ_0 .]

Tavi's note: Equation (15) is derived here and has not yet been checked numerically; test 3 of Sec. VIII A does this. It neglects the transverse waves launched at the hub and hub inertia. The transverse waves are slower by a factor $\approx \sqrt{\epsilon}$, but for the same reason their impedance is not negligible; Eq. (16) adds them, and test 3 confirms it. It is a textbook-level result for strings in the linear case; the nonlinear generalization above is Tavi's reference derivation, against which Rareş and Antonia's independent derivation is checked.

[TO DO: Finite-amplitude junction. Rareş and Antonia solve the junction exactly for an incident jump $\epsilon_p \rightarrow \epsilon_1$ of arbitrary amplitude, with all threads initially at ϵ_p : shock and rarefaction-fan relations for the cubic law, slack threads, and reduction to a single algebraic equation for the hub velocity with a proof that its root is unique. Three checks: the small-amplitude limit reproduces the nonlinear small-increment junction relations; transmission is exact for $N = 2$ and $N = 4$ at any amplitude; material D without slack threads reproduces Eq. (15). For S and D ($\epsilon_p = 0.1\epsilon_b$) they tabulate the hub strains, the incident amplitude at which thread 0 breaks at the hub, and the amplitude at which the nearest same-side neighbour goes slack, and compare them with the linear predictions. Then test 3 of Sec. VIII A is repeated at finite amplitude. Finally, they state whether the linear relations are adequate for the failure criterion and whether stiffening localizes the load.]

Reference solution (Tavi only). *Finite-amplitude junction.* Equations (15)–(20) assume small increments. The exact solution below is, like Eq. (15), the longitudinal (1D) limit: it does not include the transverse impedance of Eq. (16), which for S at $\epsilon_p = 0.1\epsilon_b$ changes the neighbour tensions by up to 15% at small amplitude. Conclusion (c) below (localization set by N , not by the material) holds within this 1D limit; with the transverse term the material enters through Z_T/Z_L (0.17 for S, 0.06 for D). In an impact the jump reaching the hub is not small: the loaded thread may be close to ϵ_b . We therefore solve the junction exactly, as a Riemann problem, for all threads initially at ϵ_p and at rest, and an incident wave in thread 0 that raises the strain from ϵ_p to ϵ_1 .

Waves in one thread. In the Lagrangian coordinate s along a thread, measured outwards from the hub, with v the particle velocity along the thread (positive outwards),

$$\partial_t \epsilon = \partial_s v, \quad \rho \partial_t v = \partial_s \sigma(\epsilon), \quad c(\epsilon) = \sqrt{\sigma'(\epsilon)/\rho} = \sqrt{(E_0 + 3b\epsilon^2)/\rho}. \quad (21)$$

For $b > 0$, c grows with ϵ : loading waves steepen into shocks and unloading waves spread into centred rarefaction fans. For $b = 0$ (material D) both keep their shape. A thread cannot carry compression, so $\sigma = 0$ for $\epsilon \leq 0$ (slack). A wave connecting a state ahead (ϵ_a, v_a) to a state behind (ϵ, v) satisfies $v - v_a = \pm \Phi(\epsilon; \epsilon_a)$, with the upper sign for waves travelling towards the hub and the lower sign for waves leaving it, where

$$\Phi(\epsilon; \epsilon_a) = \begin{cases} \sqrt{[\sigma(\epsilon) - \sigma(\epsilon_a)](\epsilon - \epsilon_a)/\rho}, & \epsilon \geq \epsilon_a \text{ (shock)}, \\ -\int_{\max(\epsilon, 0)}^{\epsilon_a} c(e) de, & \epsilon < \epsilon_a \text{ (fan)}. \end{cases} \quad (22)$$

The shock branch follows from the jump conditions $-D[\epsilon] = [v]$, $-\rho D[v] = [\sigma]$, which give the shock speed $C = \sqrt{[\sigma]/(\rho[\epsilon])}$; it contains Eq. (7) as the case $\epsilon_a = 0$, and Eq. (17) is its chord form. For the cubic law the fan integral is elementary,

$$\int c de = \frac{1}{\sqrt{\rho}} \left[\frac{e}{2} \sqrt{E_0 + 3be^2} + \frac{E_0}{2\sqrt{3b}} \operatorname{arcsinh}\left(e\sqrt{3b/E_0}\right) \right]. \quad (23)$$

Φ is continuous and strictly increasing, and both branches have slope $c(\epsilon_a)$ at $\epsilon = \epsilon_a$, so they differ only at second order; this is why Eq. (19) is recovered for small jumps. A thread can be shortened at most at the speed $v_{s1} = \int_0^{\epsilon_p} c de$; beyond this its end is slack and carries no tension.

TABLE VI. Finite-amplitude junction, $\epsilon_p = 0.1\epsilon_b$: incident amplitude ϵ_{1c} at which thread 0 breaks at the hub (exact, and from Eq. (15) applied to tensions), largest neighbour tension at that instant, and number of slack threads. Reference solution.

Material	N	ϵ_{1c}/ϵ_b (exact)	ϵ_{1c}/ϵ_b (linear)	$\sigma_{\text{neigh}}/\sigma_b$	slack threads
S	5	0.944	0.916	0.618	2
S	6	0.909	0.869	0.677	2
S	8	0.850	0.819	0.504	2
S	12	0.780	0.775	0.327	4
S	16	0.739	0.755	0.240	6
D	5	0.870	0.850	0.618	2
D	6	0.817	0.775	0.633	2
D	8	0.740	0.700	0.479	2
D	12	0.654	0.640	0.309	4
D	16	0.618	0.614	0.235	2

Hub conditions. By symmetry the hub velocity is $V\mathbf{e}_0$. The incident shock has $v_1 = \Phi(\epsilon_1; \epsilon_p)$. After the interaction the strains at the hub are ϵ_0^* in thread 0 and ϵ_j^* in thread $j \geq 1$, fixed by

$$V = v_1 - \Phi(\epsilon_0^*; \epsilon_1), \quad V \cos \varphi_j = -\Phi(\epsilon_j^*; \epsilon_p), \quad (24)$$

and equilibrium of the massless hub along $\mathbf{e}_0$ gives one equation for V ,

$$f(V) \equiv [\sigma(\epsilon_0^*) - \sigma(\epsilon_p)] + \sum_{j=1}^{N-1} [\sigma(\epsilon_j^*) - \sigma(\epsilon_p)] \cos \varphi_j = 0. \quad (25)$$

Every term decreases with V , so f is strictly decreasing; since $f(0) > 0$ the root is unique and is found by bisection. Threads with $\cos \varphi_j < 0$ receive a shock, those with $\cos \varphi_j > 0$ a fan or become slack. The wave reflected into thread 0 is a shock if $V < v_1$ ($N \geq 5$: the hub acts as a partial clamp) and a fan if $V > v_1$ ($N = 3$). If thread 0 is pre-stretched to $\epsilon_0 > \epsilon_p$ and the background is held by a constant external force, the same equations apply with $\sigma(\epsilon_p)$ replaced by $\sigma(\epsilon_0)$ in the first term of Eq. (25) and $\Phi(\epsilon_1; \epsilon_p)$ by $\Phi(\epsilon_1; \epsilon_0)$.

Checks. (i) Small jumps: for $N = 8$, $\epsilon_0 = 0.5\epsilon_b$ and a jump of 10^{-6} , $T_0/T_{\text{inc}} = 1.39422434$, identical to Eq. (19). (ii) $N = 2$ and $N = 4$: exact transmission at any amplitude, since the opposite thread receives the same shock and the threads at 90° receive nothing. (iii) Material D without slack threads coincides with Eq. (15). (iv) Direct simulation: N mass-spring chains (1500 cells per thread) joined at a massless hub solved from equilibrium at every step, with the far end of thread 0 pulled at v_1 . For S and D, $N = 3, 8, 16$ and $\epsilon_1/\epsilon_b = 0.2, 0.5, 0.8$ (18 cases), the hub strains agree with Eqs. (24)–(25) to three decimals; for example, for S, $N = 8$, $\epsilon_1 = 0.8\epsilon_b$, both give $\epsilon_0^*/\epsilon_b = 0.946, 0.639$ in the opposite thread, and a slack nearest same-side thread (scripts `ref/riemann.py`, `ref/sim.py`, `ref/tables.py`).

Validity. The state ϵ_0^* is established at $t = 0^+$, before the angles can change, so the small-hub-displacement assumption holds where the failure criterion is evaluated. The solution holds until reflections return from the next junction, $t < 2\ell_{\text{min}}/c_{\text{max}}$. At threshold the hub moves at $V \approx 130\text{--}400$ m/s (S), so the later evolution is not covered. Hub inertia and transverse waves are neglected.

Results ($\epsilon_p = 0.1\epsilon_b$). Table VI gives the incident amplitude ϵ_{1c} at which thread 0 breaks at the hub ($\epsilon_0^* = \epsilon_b$), Table VII the amplitude at which the nearest same-side neighbour goes slack, and Table VIII the tension ratios as functions of amplitude. Four conclusions follow. (a) The linear relations are adequate for the failure threshold: Eq. (15) predicts ϵ_{1c} within 6%. (b) They are not adequate for the neighbours: at $N = 8$ the opposite thread receives up to 32% more than the linear value. The cause is mainly the slackening of same-side threads, which occurs at $\epsilon_1 \approx 0.35\text{--}0.53\epsilon_b$, well below failure, and also for the linear material D. This unilateral nonlinearity is absent from the linear analysis. (c) Stiffening does not improve localization at finite amplitude: at the failure of thread 0 the largest neighbour tension is practically the same for S and D (Table VI), S being slightly worse. For $N = 5$, once the two same-side threads are slack, statics alone gives $\sigma_{\text{neigh}}/\sigma_b = 1/(2 \cos 36^\circ) \approx 0.618$ for any material. (d) For $N \geq 5$ the hub localizes: thread 0 breaks while the neighbours are at 24–68% of σ_b . For $N \leq 4$ the hub does not amplify the tension ($T_0 \leq T_{\text{inc}}$) and there is no hub-localized failure. With a pre-stretched thread 0, the deviations from Eq. (19) stay below 1% up to jumps of $0.1\epsilon_b$ without slack, and reach 6% (T_0) and 13% (opposite neighbour) once slack occurs ($\epsilon_0 = 0.67\epsilon_b$, jump $0.2\epsilon_b$).

TABLE VII. Incident amplitude ϵ_1/ϵ_b at which the nearest same-side neighbour ($j = 1$) goes slack, $\epsilon_p = 0.1\epsilon_b$: exact value, with the linear prediction in parentheses. Reference solution.

N	S	D
5	0.448 (0.430)	0.505 (0.505)
6	0.365 (0.358)	0.400 (0.400)
8	0.347 (0.345)	0.383 (0.383)
12	0.385 (0.391)	0.446 (0.446)
16	0.435 (0.448)	0.533 (0.533)

TABLE VIII. Finite-amplitude junction, material S, $\epsilon_p = 0.1\epsilon_b$: tension ratios at the hub as functions of the incident amplitude. Linear values, Eq. (15): 1.500 and 0.500 for $N = 8$; 1.750 and 0.250 for $N = 16$. Reference solution.

N	ϵ_1/ϵ_b	$\Delta T_0/T_{\text{inc}}$	$\Delta T_{\text{opp}}/T_{\text{inc}}$	slack threads
8	0.2	1.508	0.505	0
8	0.3	1.519	0.513	0
8	0.4	1.492	0.553	2
8	0.6	1.428	0.625	2
8	0.8	1.397	0.659	2
16	0.2	1.775	0.254	0
16	0.4	1.854	0.268	0
16	0.6	1.847	0.327	4
16	0.8	1.813	0.376	6

B. Dynamic load transfer at a sudden break

In the quasi-static regime the failure of a thread is still a fast event: the release wave travels at c_L and the load is transferred to the remaining threads in a time much shorter than the period of the net–drone system. The overshoot follows from energy conservation. If a structure with restoring force $F(w)$ is suddenly subjected to a constant load P starting from w_0 at rest, the maximum deflection w_m satisfies

$$\int_{w_0}^{w_m} F(w) dw = P(w_m - w_0). \quad (26)$$

For $F = kw^n$, $w_0 = 0$ and $P = F(w_s)$, Eq. (26) gives

$$\frac{w_m}{w_s} = (n+1)^{1/n}, \quad \frac{F(w_m)}{P} = n+1. \quad (27)$$

The linear case gives the classical factor 2. For a stiffening response ($n > 1$) the *deflection* overshoot is smaller (1.59 for $n = 3$, 1.35 for $n = 7$), while the *force* overshoot is larger. Because threads fail at a critical *strain*, stiffening protects the remaining threads.

In a net the effective exponent combines material and geometric stiffening. For an unstressed star, $\epsilon \propto w^2$ and $F \propto T(\epsilon)w$, so $n_{\text{eff}} = 3$ for a linear material. The cubic law adds material stiffening, but it does not make n_{eff} equal to 7: that would require the cubic term to dominate $T(\epsilon)$, and for S the linear term still dominates at $\epsilon = 0.8\epsilon_b$ (local material exponent $d \ln T / d \ln \epsilon = 1.78$). The measured value at the operating point is $n_{\text{eff}} = d \ln F / d \ln w = 3.87$ for S and 2.97 for D. The strain overshoot is $(w_m/w_s)^2$. As a benchmark, when one of N radials is lost while the others carry a constant load P , Eq. (26) with $F \rightarrow \frac{N-1}{N}F$ gives the values of Table IX.

Numerical protocol. The numerical column of Table IX is obtained with a constant vertical force P applied to a hub of point mass m_{hub} (a light node, not a drone-held node). The static state is reached with viscous damping, which is then switched off; at t_b all segments of radial 0 are removed and the maximum strain ϵ_m in the remaining radials is recorded. P is chosen so that the static strain is ϵ_0 in a linear regime (material D) and in a cubic-dominated regime (material S, $\epsilon_0 \approx 0.8\epsilon_b$), and $n_{\text{eff}} = d \ln F / d \ln w$ is measured at the operating point. The same set-up with the tension of radial 0 decreased linearly to zero over a time t_f gives the dependence on the release time (Fig. 2).

[TO DO: Rares and Antonia (i) derive Eqs. (26)–(27); (ii) redo Table IX; (iii) treat the relevant case for impact, in which the drone is still moving when the thread breaks: the remaining net must absorb

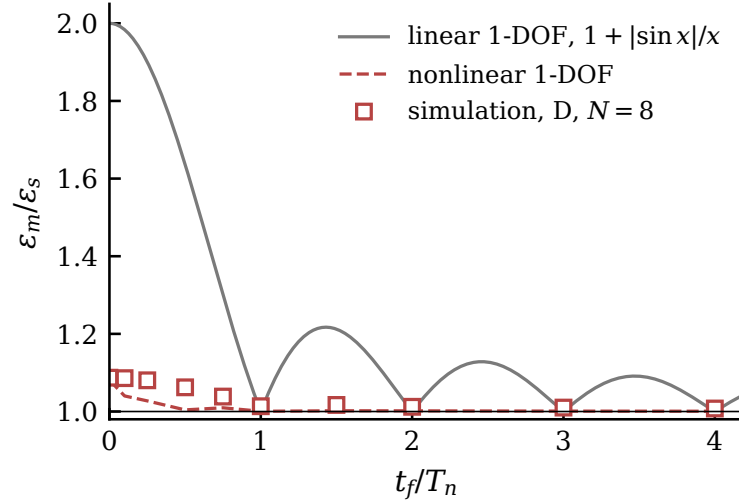


FIG. 2. Dynamic overload after the loss of one of $N = 8$ radials as a function of the release time t_f , normalized by the period T_n of the hub mode: maximum strain in the remaining radials over its static value after the loss, material D, hub constrained to move along z . Markers: Algorithm 1; line: linear single-degree-of-freedom response to a ramp, $1 + |\sin x|/x$ with $x = \pi t_f/T_n$. The geometric stiffening of the net ($n_{\text{eff}} \approx 3$) reduces the instantaneous overshoot from 2 to 1.09, and the overshoot is below 2% for $t_f \gtrsim T_n$. Dashed: nonlinear single-degree-of-freedom response.

TABLE IX. Loss of one of N radials under a constant load, power-law response $F \propto w^{n_{\text{eff}}}$; strain ratio ϵ_m/ϵ_0 after the break, static and dynamic, from Eq. (26). Numerical values from Algorithm 1, hub constrained to move along z . For S the power law is only an approximation (the material exponent varies with strain), hence the 1–3% difference. With the hub free in the plane it drifts sideways after the break and the overshoot is much larger (D, $N = 8$: 1.754); this is a separate mechanism and is discussed in the text.

N	n_{eff}	static ϵ_s/ϵ_0	dynamic ϵ_m/ϵ_0 (analytical)	dynamic (numerical)
4	3 (D)	1.211	1.429	1.432
4	3.87 (S)	1.160	1.321	1.279
8	3 (D)	1.093	1.188	1.189
8	3.87 (S)	1.072	1.143	1.131
16	3 (D)	1.044	1.088	1.089
16	3.87 (S)	1.034	1.068	1.063

$\frac{1}{2}Mv_r^2$ plus the energy released by the broken thread's share of the load; (iv) include the finite failure time, following [12].]

Reference solution (Tavi only). (i) A body loaded by the structure with restoring force $F(w)$ and by a constant force P starts from rest at w_0 and stops momentarily at w_m ; the work–energy theorem between these two states gives Eq. (26). For $F = kw^n$, $w_0 = 0$, $P = kw_s^n$: $kw_m^{n+1}/(n+1) = kw_s^n w_m$, so $(w_m/w_s)^n = n+1$ and $F(w_m)/P = (w_m/w_s)^n = n+1$, Eq. (27). (ii) Before the break $P = kw_0^n$; after it $F \rightarrow \frac{N-1}{N}kw^n$. Static: $r_s = (N/(N-1))^{1/n}$. Dynamic, from Eq. (26) with $r = w_m/w_0$: $\frac{N-1}{N} \frac{r^{n+1}-1}{n+1} = r-1$ (Appendix C). For the unstressed star $\epsilon \simeq w^2/2R^2$, so $\epsilon/\epsilon_0 = r^2$. All analytical entries of Table IX are reproduced to three decimals. (iii) *Drone case*. There is no constant load: the drone is a mass with kinetic energy. When a segment fails at w_1 with drone speed v_r , $\frac{1}{2}Mv_r^2 = E_{\text{kin}} - W(w_1)$, and the arrest condition for the damaged net is $\int_{w_1}^{w_2} F_{\text{rem}} dw \geq \frac{1}{2}Mv_r^2$, with w_2 the next failure. The strain energy of the broken segment, $\rho A \ell e_{\text{mat}}$ per unit mass, is not transferred to the rest of the net: it is radiated by the release wave and dissipated. In the massless-thread limit the strains of the surviving threads do *not* jump at the break when the failed segment ends at the node held by the drone: for the off-centre star computed above, the hub equilibrium and all surviving strains are identical just before and just after the failure of the outer segment. The overshoot of Table IX is therefore not the relevant mechanism for the drone impact on a star; it applies when a load is sustained (e.g. the weight of the arrested drone in a horizontal net) or when the break occurs at a light node (hub, ring junction) not in contact with the drone. (iv) *Finite failure time*. For a linear single-degree-of-freedom system of period T_n , if the load

of the failed member is transferred linearly in a time t_f , the response to the ramp gives

$$\text{DAF}(t_f) = 1 + \frac{|\sin(\pi t_f/T_n)|}{\pi t_f/T_n},$$

which tends to 2 for $t_f \ll T_n$ and to 1 for $t_f \gg T_n$, qualitatively consistent with the role of the failure time studied in [12] (to be checked against their results). For the net, T_n is the period of the local mode of the node that loses the member; in the massless-thread limit $T_n \rightarrow 0$ for drone-held nodes, which is consistent with (iii). For the nonlinear law the ramp response must be integrated numerically.

Tavi’s note: Consequence of reference solution (iii) for the paper: in the configuration of Fig. 1c the dynamic overload mechanism of Secs. V A–V B does not operate, because the broken segment ends at the node carried by the drone. The dynamic localization criterion becomes relevant only for breaks at light nodes, i.e. with rings or with contact spread over several threads. This must be decided before Rareş and Antonia write Sec. V C.

Tavi’s note: The “static load” case of Table IX is a benchmark, not the impact case. For S the effective exponent changes with strain (from linear at small ϵ to cubic near ϵ_b), so Rareş and Antonia should use the full law rather than a single n . The comparison with Hedgepeth’s factors 1.15–1.27 for composites [10] and with the code value 2.0 for cable loss [11] belongs in the Discussion.

C. Localization criterion

Let a thread (or segment) fail at time t_b , while its neighbours have strain $\epsilon_n(t_b)$. We define the localization number

$$\Lambda_{\text{loc}} = \frac{\epsilon_n(t_b) + \Delta\epsilon_{\text{dyn}}}{\epsilon_b}, \quad (28)$$

where $\Delta\epsilon_{\text{dyn}}$ is the dynamic strain increase of the most loaded neighbour caused by the break (Secs. V A–V B). Failure stays localized if $\Lambda_{\text{loc}} < 1$ and cascades if $\Lambda_{\text{loc}} \geq 1$. Stiffening lowers Λ_{loc} in two ways: through the smaller strain overshoot [Eq. (27)] and through the impedance mismatch at junctions (Sec. V A).

[TO DO: Rareş and Antonia express Λ_{loc} in terms of the groups of Eq. (6) and of N for the off-centre impact on the star net and on the star with one ring, and determine the boundary $\Lambda_{\text{loc}} = 1$ for S and D (Fig. 6). In the limit of slow loading, they compare the predicted number of broken threads with Table S5 of [1].]

Reference solution (Tavi only). *Strain form, star net, off-centre impact.* From the minimization of the off-centre model, just before (and just after) the failure of the outer segment the most loaded neighbour is the inner segment, with $\epsilon_n/\epsilon_b = 0.88, 0.58, 0.37, 0.22$ for $a/R = 0.2, 0.3, 0.4, 0.5$ (S, $N = 8$, $\epsilon_p = 0$; D gives 0.89, 0.55, 0.33, 0.19), and the largest radial strain is about half of that. With $\Delta\epsilon_{\text{dyn}} = 0$ [reference solution (iii) above], $\Lambda_{\text{loc}} = \epsilon_n/\epsilon_b < 1$: the first break never cascades *instantaneously*. *Energy form.* Whether failure stays localized is then decided by the remaining kinetic energy. With $E_{\text{kin}}/(m_{\text{net}}e_{\text{mat}}) = \Pi_E M/m_{\text{net}}$,

$$\Lambda_E(a) = \frac{\Pi_E M/m_{\text{net}} - \eta_A(a)}{\eta_B(a) - \eta_A(a)},$$

where $\eta_{A,B}(a; N, \epsilon_p/\epsilon_b, b\epsilon_b^2/E_0, \epsilon_b)$ are the utilizations of the reference solution of Sec. IV B. $\Lambda_E \leq 0$: no failure; $0 < \Lambda_E < 1$: arrested with one sacrificial break (criterion B); $\Lambda_E \geq 1$: a second failure occurs. The boundary $\Lambda_E = 1$ is $m_{\text{net}} = \Pi_E M/\eta_B(a)$, i.e. exactly m_{min}^B at that impact point, so the localization criterion and the mass saving of Eq. (29) are the same statement. In the plane $(a/R, \Pi_E M/m_{\text{net}})$ the boundary is the curve $\eta_B(a)$ of the table above, nearly flat (0.24–0.26 for S, 0.23–0.30 for D). *Comparison with Cranford et al.* No reference values: the QFM prediction of their Sec. S10 is written for a power-law exponent fitted to each material, which Rareş and Antonia must fit to Eq. (2) first; the numbers of their Table S5 should be read from the PDF, as the text extraction is not reliable. *Star with one ring.* No closed form is given here: the ring adds two junctions on the impacted radial and the off-centre model of Sec. IV B has to be extended with the ring nodes as unknowns. The simulations fix what the analysis must reproduce: at m_{min}^B , for S and D, only the outer half of the impacted radial (between the ring junction and the frame) fails, and the ring itself does not break (Fig. 6c,d). The localization boundary for the ring is therefore again the first sacrificial break of a single thread piece, and Λ_E applies with $\eta_{A,B}$ of the ring net. *Dynamic map.* Figure 9 gives the map in the variables $(a/R, m_{\text{net}}/m^*)$, with $m^* = E_{\text{kin}}/e_{\text{mat}}$, i.e. $m_{\text{net}}/m^* = 1/(\Pi_E M/m_{\text{net}})$. The

boundary $\Lambda_E = 1$ is the lower edge of the arrested region, $m_{\min}^B(a)/m^*$, and the onset of failure is $m_{\min}^A(a)/m^*$. For S, $m_{\min}^{\text{Bany}}/m^* = 3.7\text{--}3.9$ at $a = R/4\text{--}R/2$, slightly below the drone-held $1/\eta_B = 4.0$, while $m_{\min}^A/m^*$ is far below the drone-held $1/\eta_A$ (8.1 against 13.4 at $R/2$): the drone moves in the plane, and $m_{\min}^A/m^*$ is close to $1/\eta_A$ of the free limit at $R/4$ (4.5 against 4.4) and 20% above it at $R/2$ (8.1 against 6.7). For D the arrested-with-failure region is almost absent: $m_{\min}^B$ and $m_{\min}^A$ nearly coincide, and below them the runs perforate or time out. The isolated arrested runs of D at $a \geq 0.5R$ and $m_{\text{net}} \leq 2.5m^*$ are not captures: after the first break the drone keeps 78–87% of E_{kin} as in-plane velocity, and its normal velocity reverses while it is thrown sideways (the sling of Sec. VIII).

Tavi’s note: This is the main physics contribution. The quasi-static quantized-fracture-mechanics model of Pugno in the Supplementary Information of [1] (Sec. S10) already predicts localization versus a power-law exponent; Rareş and Antonia must read it and position their criterion as its dynamic, discrete-net extension for the cubic law. The claim of novelty must be checked against the literature on dynamic overload in nonlinear cable nets before submission.

D. Mass saving due to sacrificial failure

Under criterion B, the sacrificial segment absorbs its full energy $\rho A \ell e_{\text{mat}}$ before failing, and the remaining net continues to absorb energy. The utilization becomes $\eta_B(a) \geq \eta_A(a)$ and

$$\frac{m_{\min}^A}{m_{\min}^B} = \frac{\min_a \eta_B(a)}{\min_a \eta_A(a)}. \quad (29)$$

[TO DO: Rareş and Antonia estimate Eq. (29) analytically for the star net (one sacrificial outer segment, the remaining star as in Sec. IV), for S and D, and compare with the simulations (Table XIV, Fig. 7). Expected: a saving for S, and little or none for D, where the break cascades.]

Reference solution (Tavi only). With the off-centre model and $\mathcal{P}$ the disc of radius $R/2$, the worst case of both criteria is at the edge $a = R/2$ (the minimum over the sampled points $a/R = 0.05\text{--}0.5$). For $N = 8$:

	$\epsilon_p = 0$			$\epsilon_p = 0.1\epsilon_b$		
	$\min \eta_A$	$\min \eta_B$	$m_{\min}^A/m_{\min}^B$	$\min \eta_A$	$\min \eta_B$	$m_{\min}^A/m_{\min}^B$
S	0.068	0.244	3.6	0.075	0.251	3.3
D	0.069	0.231	3.3	0.078	0.237	3.0

The saving is large and almost the same for S and D. This contradicts the expectation stated in the TODO (“little or none for D”): in the star net the second failure is the inner segment of the same radial, not a neighbour, for both materials, and η is controlled by geometry. The material effect can only appear in configurations where the break loads other threads through a light node (ring junctions) or where the drone contacts several threads. The absolute values are also low ($m_{\min}^B \approx 4m_{\min}^{\eta=1}$), because at $a = R/2$ a single radial carries the drone: a star alone is a poor design for off-centre impacts, and $m_{\min}^A \propto N$ at fixed a . *Caveats from the simulations.* This estimate holds the gripped point at $(a, 0)$ and corresponds to criterion B_{any} (the second failure is not required to be near the drone). With the drone free in the plane, $\eta_A(R/2)$ doubles (0.136 for S, 0.180 for D at $\epsilon_p = 0$, Sec. IV B). Test 5 with the gripped point free in the plane also after the first break gives $\eta_B(R/2) = 0.30$ (S) and 0.60 (D) at $\epsilon_p = 0.1\epsilon_b$; the gripped point slides to the hub before the second failure. The free-limit estimate of Eq. (29) at the worst point is then $0.30/0.149 = 2.0$ for S and $0.60/0.186 = 3.2$ for D. *Comparison with the simulations.* The dynamic saving is 2.0–2.1 for S and 1.01–1.05 for D (Table XIV). For S it equals the free-limit estimate. For D it is three times smaller: the expectation of the TODO holds dynamically, although neither quasi-static estimate shows it, and the reason is the sling described in Sec. VIII. With the drone constrained to move along z , the dynamic saving is 3.46 (S) and 3.18 (D), close to the drone-held estimates 3.3 and 3.0. For S, the dynamic $\eta_A(R/2) = 0.128$ lies between the held and free limits; for D it is 0.192, at the free limit. At $m_{\min}^B$ only the impacted radial fails, for both nets and materials (Fig. 6).

VI. NUMERICAL METHOD

Tavi’s note: This section is Tavi’s: he writes the simulator. Rareş and Antonia’s role is validation (Sec. VIIIA).

A. Discretization

Each thread is discretized into n_s segments; nodes carry lumped masses $m_i = \sum_{e \ni i} \frac{1}{2} \rho A_e \ell_e / n_s$, anchored nodes are fixed. Each segment is a nonlinear spring that carries tension only (no compression, no bending):

$$\mathbf{f}_{ij} = T(\epsilon_{ij}) \frac{\mathbf{x}_j - \mathbf{x}_i}{\|\mathbf{x}_j - \mathbf{x}_i\|}, \quad \epsilon_{ij} = \frac{\|\mathbf{x}_j - \mathbf{x}_i\|}{\ell_{ij}} - 1, \quad T = 0 \text{ if } \epsilon_{ij} \leq 0. \quad (30)$$

A segment is removed permanently when $\epsilon_{ij} \geq \epsilon_b$ (several segments may fail in the same step); the time and location of each failure are recorded, and the elastic energy of the failed segment, $A_e(\ell_e/n_s)\Phi(\epsilon_{ij})$, is booked as dissipated. The prestressed initial state is obtained from Eq. (4); the segment nodes are placed on the straight line between the end nodes of each thread, so the initial state is an exact equilibrium.

Contact. The drone is a rigid sphere of mass M and radius r_d . Following Sec. II, at first contact the net node closest to $\mathbf{p}$ is attached to the drone (gripped contact): it follows the drone, its mass is added to M , and the resultant segment force on it acts on the drone. The capture is inelastic and dissipates $\Delta E_{\text{cap}} = \frac{1}{2} \frac{Mm}{M+m} v_{\text{rel}}^2$, with m the captured node mass; this term is included in the energy balance. All other net nodes interact with the drone through a frictionless Hertz-type penalty force $\mathbf{f}_c = k_c \delta^{3/2} \hat{\mathbf{n}}$ for penetration depth $\delta > 0$, with $\hat{\mathbf{n}}$ the outward normal of the sphere. For material D the contact stiffness is set relative to the axial stiffness of a segment, $k_c = f_c^2 (E_0 A_e n_s / \ell_e) / (\frac{3}{2} \delta_{\text{ref}}^{1/2})$, so that the contact frequency at the reference penetration δ_{ref} is $f_c = 8$ times the axial frequency of a segment; for the softer material S a fixed $k_c = 10^7 \text{ N/m}^{3/2}$ is used. The reference penetration is $\delta_{\text{ref}} = 0.01 r_d = 1.5 \text{ mm}$, and the run is stopped if δ exceeds it (Table XVI).

Energy balance. At every output time the kinetic energies of the drone and of the net, the elastic energy of the intact segments, the energy dissipated by failure, the capture loss, and the contact energy $\frac{2}{5} k_c \delta^{5/2}$ are recorded. Their sum is conserved by the scheme to second order in Δt ; the relative error at the end of the run, normalized by $E_{\text{kin}} + U_{\text{prestress}}$, is reported for every run. The utilization of a run is

$$\eta = \frac{U_{\text{el}} + W_{\text{fail}} - U_{\text{prestress}}}{m_{\text{net}} e_{\text{mat}}}, \quad (31)$$

evaluated at arrest, where U_{el} is the elastic energy of the intact segments and W_{fail} the energy dissipated by failure. No damping is used in the impact runs; viscous damping is used only to reach static equilibrium in the quasi-static validations and in the set-up of the overload test (Sec. VB).

B. Time integration

We use explicit velocity Verlet with a fixed step

$$\Delta t = \min \left(C \frac{\min_e \ell_e / n_s}{c_{\text{tan}}}, \frac{2C}{\omega_c} \right), \quad c_{\text{tan}} = \max_{0 \leq \epsilon \leq \epsilon_b} \sqrt{\frac{T'(\epsilon)}{\mu_0}}, \quad C \leq 0.5. \quad (32)$$

The stability limit is set by the tangent wave speed, not by the chord speed c_L of Eq. (7): for S, $c_{\text{tan}}(\epsilon_b) \approx 2480 \text{ m/s}$, against $c_L(\epsilon_b) = 1754 \text{ m/s}$. The second limit is set by the contact frequency $\omega_c = \sqrt{k_{\text{eff}}/m_{\text{min}}}$, with $k_{\text{eff}} = \frac{3}{2} k_c \delta_{\text{ref}}^{1/2}$ and m_{min} the smallest node mass. For D with 1 cm segments, $\Delta t \approx 0.5 \mu\text{s}$; an arrest of $\sim 15 \text{ ms}$ takes $\sim 3 \times 10^4$ steps.

The simulator also supports kinematic forcing (a prescribed velocity or displacement of a node, used in validation tests 1–3 and 5) and a constant force on a node (Sec. VB). The procedure is summarized in Algorithm 1; Algorithm 2 finds the minimum mass for a given topology and criterion.

Algorithm 1 Explicit mass–spring simulation of a drone impact on a prestressed net with thread failure

Require: net $(G, \mathcal{A}, q, A)$, material law $T(\epsilon)$, ϵ_b ; drone $(M, r_d, v_0, \mathbf{p})$; n_s , C , t_{end}
Ensure: trajectory $\{\mathbf{x}^n\}$, list of failures $\{(e, t_b)\}$, energy balance, outcome (arrested / perforated / timeout)

- 1: solve Eq. (4) for the prestressed geometry; compute rest lengths ℓ_e
- 2: subdivide threads into segments; lump masses
- 3: place drone tangent to the net at $\mathbf{p}$, velocity $-v_0 \hat{\mathbf{z}}$; set Δt from Eq. (32)
- 4: **while** $t < t_{\text{end}}$ **do**
- 5: update positions: $\mathbf{x} \leftarrow \mathbf{x} + \mathbf{v} \Delta t + \frac{1}{2} \mathbf{a} \Delta t^2$ (anchors fixed, gripped node follows drone)
- 6: compute segment strains ϵ_{ij}
- 7: **for** each segment with $\epsilon_{ij} \geq \epsilon_b$ **do**
- 8: remove segment; append (e, t) to failure list; book its energy as dissipated ▷ sacrificial failure
- 9: **end for**
- 10: tensions $T(\epsilon_{ij})$ (zero if slack); contact forces; capture of the gripped node at first contact
- 11: new accelerations $\mathbf{a}'$; update velocities: $\mathbf{v} \leftarrow \mathbf{v} + \frac{1}{2} (\mathbf{a} + \mathbf{a}') \Delta t$
- 12: save $(t, \mathbf{x}, \mathbf{v})$, intact mask, energy balance) at fixed output interval ▷ HDF5, ML part
- 13: **if** penetration $\delta > \delta_{\text{ref}}$ **then return** error (time step not valid)
- 14: **end if**
- 15: **if** drone vertical velocity changes sign **then return** arrested, w_{max} ▷ end of the capture stroke
- 16: **end if**
- 17: **if** drone passes the anchor plane by more than $2R$ **then return** perforated
- 18: **end if**
- 19: **end while**
- 20: **return** timeout

Algorithm 2 Minimum mass for a topology and an arrest criterion

Require: topology $(G, \mathcal{A}, q)$, material, drone (M, v_0) , criterion $X \in \{A, B\}$, impact points $\{\mathbf{p}_k\} \subset \mathcal{P}$

- 1: $s_0 \leftarrow$ scale with $m_{\text{net}} = E_{\text{kin}}/e_{\text{mat}}$ ▷ sections scaled uniformly, $A_e = s \hat{A}_e$
- 2: $s_{\text{lo}} \leftarrow 0$; $s_{\text{hi}} \leftarrow s_0$; **while** X fails for some $\mathbf{p}_k$ **do** $s_{\text{lo}} \leftarrow s_{\text{hi}}$, $s_{\text{hi}} \leftarrow 2s_{\text{hi}}$
- 3: **while** $(s_{\text{hi}} - s_{\text{lo}})/s_{\text{hi}} > \delta$ **do**
- 4: $s \leftarrow (s_{\text{lo}} + s_{\text{hi}})/2$
- 5: run Algorithm 1 for every $\mathbf{p}_k$
- 6: **if** criterion X holds for all $\mathbf{p}_k$ **then** $s_{\text{hi}} \leftarrow s$
- 7: **else** $s_{\text{lo}} \leftarrow s$
- 8: **end if**
- 9: **end while**
- 10: **return** $m_{\text{min}}^X = \sum_e \rho s_{\text{hi}} \hat{A}_e \ell_e$, worst-case impact point, broken-thread map

Tavi’s note: Criterion A: no failure recorded and arrested. Criterion B_{any}: arrested, not perforated, and fewer than k_{max} failed threads. Criterion B_{loc}: in addition, the first failure of every failed thread within R_{max} of the drone in-plane position at that time; reported for $R_{\text{max}}/R = 0.25, 0.5, 0.75$. A run that is not arrested within the simulated time $t_{\text{end}} = 0.25$ s (timeout) fails every criterion. This is conservative and does not change the results: with t_{end} doubled, 13 of the 18 timeouts in which the drone was still moving end in perforation and none in arrest, and the 10 timeouts in which the drone was slow and still inside the frame, rerun to 0.5 s, all remain timeouts. Failures are counted per thread and not per segment: a broken radial typically fails at several segments (2–6 at $n_s = 40$, up to 10 at $n_s = 80$), so a count of segments makes criterion B depend on the discretization; with the thread count, $m_{\text{min}}^{\text{Bany}}$ is identical at $n_s = 40$ and 80. All criteria use the same relative tolerance $\delta = 1\%$ and the same runs, so that $m_{\text{min}}^A \geq m_{\text{min}}^{\text{Bloc}} \geq m_{\text{min}}^{\text{Bany}}$ holds by construction; this ordering is checked on every row of Table XIV. The worst-case impact point is not the same for all criteria (for S, $a = R/2$ for A and $a = R/4$ for B_{any}). Uniform scaling of sections is the simplest choice; optimizing sections per edge belongs to the ML part. Criterion A is monotone in s ; criterion B is not guaranteed to be (a heavier net may fail in a different pattern), so a scan of 24 values of s precedes the bisection. Repeating the bisection from a different starting bracket reproduces m_{min}^A and $m_{\text{min}}^{\text{Bany}}$ within 2%, and $m_{\text{min}}^{\text{Bloc}}$ within 9% (star, S, $v_0 = 20$ m/s).

VII. SPECTRAL DESCRIPTORS AND LEARNED SURROGATE

Tavi’s note: This section belongs to the ML part (Octavian). Only the structure and the minimum content expected by a physics audience are fixed here.

TABLE X. HDF5 layout: one file per run (written by Algorithm 1).

Dataset / attribute	Shape	Content
/params (attributes)	—	material (E_0, b, ρ, ϵ_b), $\epsilon_p, R, M, r_d, v_0, \mathbf{p}$, criterion, contact mode, $n_s, C, \Delta t, k_c$, code version
/graph/nodes	$(n_v, 2)$	node positions of the prestressed net (m)
/graph/edges	$(n_e, 2)$	thread connectivity
/graph/anchored	(n_v)	anchored flag
/graph/q, /graph/A	(n_e)	force densities (N/m), sections (m^2)
/t	(n_t)	output times (s)
/x, /v	$(n_t, n_{\text{seg nodes}}, 3)$	node positions (m) and velocities (m/s)
/intact	(n_t, n_{seg})	intact mask per segment
/drone	$(n_t, 6)$	drone position and velocity
/seg/edges	$(n_{\text{seg}}, 2)$	segment connectivity
/seg/parent, /seg/rest_length	(n_{seg})	parent thread, segment rest length
/failures	$(n_f, 3)$	segment index, parent thread, failure time
/energy	$(n_t, 6)$	kinetic (drone, net), elastic, failure, capture, contact energy
/outcome (attributes)	—	outcome, w_{max}, η [Eq. (31)], R_d , cascade flag, energy error

TABLE XI. Data splits. Generalization is tested on topologies not seen during training.

Split	Topology families	Material	v_0 (m/s)	Topologies	Runs
Train	?	?	?	?	?
Validation	?	?	?	?	?
Test (interpolation)	?	?	?	?	?
Test (unseen topologies)	?	?	?	?	?

A. Data set

The training data are the runs of Algorithms 1–2 over a family of topologies (star, star with rings, generated planar graphs), stored in the format of Table X. **[TO DO: Number of topologies and runs, parameter ranges, impact points per topology, saved frames.]**

B. Spectral descriptors (go/no-go test)

For each net we compute the first k generalized eigenvalues of Eq. (5) and the localization of the low-frequency modes (participation ratio). We test whether these descriptors predict (i) η and (ii) whether failure stays localized, against a GNN that sees the full graph. **[TO DO: Octavian: descriptor set, regression model, criterion (e.g. spectral descriptors explain at least $\sim 80\%$ of the variance explained by the GNN).]**

C. Graph representation, architecture, and training

Nodes of the discretized net are graph nodes (features: position, velocity history, anchored flag, contact flag); edges are segments (features: rest length, section, current strain, intact flag). **[TO DO: Octavian: encoder–processor–decoder architecture [22, 23]; targets: nodal accelerations and per-edge failure probability; loss; training details.]**

D. Evaluation metrics

- rollout error on node positions and drone trajectory;

- arrested/perforated accuracy; per-edge failure precision and recall; error on R_d and on η ;
- consistency with the analytical results: η_A of Eq. (13) for hub impacts, the junction coefficients of Table IV, the localization boundary of Sec. VC;
- wall-clock speed-up relative to Algorithm 1.

VIII. RESULTS

[TO DO: Short paragraph introducing the three parts: validation of the simulator, analytical versus numerical comparison, and the learned surrogate.]

A. Validation of the simulator

1. **Test 1: single thread.** A long thread hit transversely at constant v_0 . Compare c_L , c_T , the strain ϵ and T_1 with Eqs. (7)–(9), for S and D, below the critical velocity of Eq. (10) (Table XII). The strain agrees within 0.1% and the kink speed within 0.8%. With the front speed measured from the slope of the front position, the longitudinal front agrees within 0.07% for D and 0.03% for S at 100 m/s; for S at 300 m/s, where the front is a finite-amplitude shock of the stiffening material, the difference is 1.1%. For S at 500 m/s the strain differs by 1.9% and the longitudinal front by 3.0% at 800 segments; the strain error decreases with resolution (6.6%, 4.3%, 1.9% at 200, 400, 800 segments), i.e. the strong shock of the stiffening material converges slowly. Figure 3 shows the strain profiles and thread shapes for S at 300 m/s and D at 500 m/s.
2. **Test 2: hub impact on a star net.** $F(w)$ and η_A against Eqs. (12)–(13). The stopping distance agrees to four digits and η_A within 0.5% (Table III, Fig. 4).
3. **Test 3: junction.** A single tension pulse sent along one radial of an N -star; transmitted and reflected amplitudes against Eqs. (15) and (16) (Table IV). The test is repeated with thread 0 pre-stretched to ϵ_0 , where the hub tension agrees with the small-increment nonlinear relation within 0.4% for $\epsilon_0 \leq 0.67\epsilon_b$ (Table V), and with a finite incident jump. At finite amplitude the simulated hub tension of S agrees with the small-amplitude two-dimensional value within 1% for $\epsilon_1 \leq 0.3\epsilon_b$ ($N = 8$: 1.573 and 1.553 against 1.572; $N = 16$: 1.802 and 1.799 against 1.786), and the amplitude at which thread 0 breaks at the hub is 0.852, 0.768 and 0.711 ϵ_b for $N = 8, 12$ and 16 (exact longitudinal solution: 0.850, 0.780, 0.739). Above $0.3\epsilon_b$, for $N \leq 6$, for material D, and for the slackening of the neighbours, the hub state could not be measured reliably in the net simulation (the tension behind the shock overshoots its plateau, and the transverse front reaches the hub within the measurement window), and these cases are not used.
4. **Test 4: convergence** in the number of segments per thread, n_s (Table XIII): $w_{\max}$ and η change by less than 0.1% between $n_s = 40$ and 80. For the minimum masses of Algorithm 2 (star, $M = 1$ kg, $v_0 = 15$ m/s), $m_{\min}^A$ and $m_{\min}^{\text{Bany}}$ are identical at $n_s = 40$ and 80 (same final bisection interval); at $n_s = 10$ they are 6.7% and 7.0% higher for S and 1.2% and 2.6% different for D. The star results of Table XIV therefore use $n_s = 40$; the star with ring uses $n_s = 10$. For the star with ring at $v_0 = 20$ m/s, $n_s = 20$ changes $m_{\min}^A$ by at most 1.6%, $m_{\min}^{\text{Bany}}$ by 0.7%, and the saving by 2.2%; at $a = 0.8R$, $n_s = 10, 20$ and 40 give $m_{\min}^A = 24.56, 24.80$ and 24.56 g (S, $v_0 = 10$ m/s).
5. **Test 5: off-centre impact, quasi-static.** Slow prescribed displacement of the gripped point at distance a on radial 0; $\eta_A(a)$, $\eta_B(a)$, and the first failing segment (inner or outer) against the energy-minimization model of Sec. IVB: η_A within 0.5% with the drone held and free in the plane, η_B within 1.5% with the drone held (Fig. 5a,b). With the gripped point also free after the first break, $\eta_B(R/2) = 0.30$ for S and 0.60 for D; the gripped point slides to the hub before the second failure.
6. **Test 6: energy conservation.** Reference case without failure; relative energy error for $C = 0.5, 0.25, 0.125$. With frictionless contact the error is 2.1×10^{-5} , 5.8×10^{-6} , 1.4×10^{-6} , i.e. second order in Δt . With gripped contact it is 1.9×10^{-5} , 2.5×10^{-6} , 4.8×10^{-7} , also at least second order, after the capture bookkeeping was corrected to use the velocity of the centre of mass of drone and captured node.

TABLE XII. Test 1: single-thread transverse impact. Analytical values from Eqs. (7)–(9); numerical values from Algorithm 1 with 800 segments.

Material	v_0 (m/s)	ϵ an.	ϵ num.	err. (%)	c_L an. (m/s)	c_L num.	err. (%)	c_T an. (m/s)	c_T num.	err. (%)
S	100	0.02285	0.02285	<0.001	1244	1244	0.03	185.9	185.5	0.24
S	300	0.09601	0.09611	0.10	1302	1316	1.07	385.4	385.2	0.06
S	500	0.1702	0.1669	1.92	1426	1468	2.98	543.8	544.6	0.14
D	100	0.001267	0.001267	0.01	10622	10615	0.07	377.8	374.8	0.80
D	500	0.01106	0.01106	0.002	10622	10615	0.07	1110.8	1107.1	0.33
D	900	0.02454	0.02454	0.004	10622	10614	0.07	1643.8	1640.1	0.23

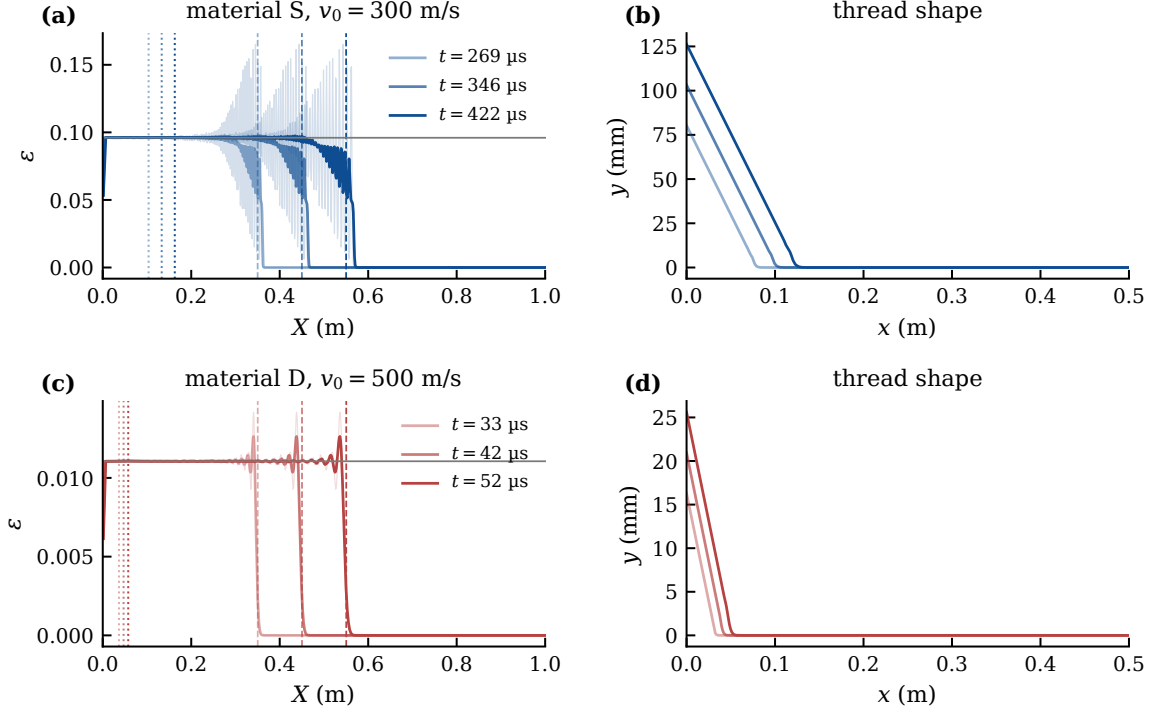


FIG. 3. Test 1: transverse impact on a single thread, (a,b) material S at $v_0 = 300$ m/s and (c,d) material D at $v_0 = 500$ m/s, at three times (light to dark). (a,c) Strain along the Lagrangian coordinate X measured from the impact point; dashed lines: $c_L t$, dotted lines: $c_T t$, from Eqs. (7)–(8); grey line: Smith strain, Eq. (9). Thin lines: nodal strain, which oscillates at the grid scale behind the front (no artificial viscosity), strongly for the shock of the stiffening material S; thick lines: 9-node moving average. (b,d) Thread shape: the straight inclined segment between the impact point and the kink.

B. Analytical versus numerical comparison

Impact regime. At fixed v_0 , the minimum mass is proportional to the drone mass, $m_{\min} \propto M$, for all criteria (within the 1% bisection tolerance for $M = 0.25, 1$ and 2 kg). It is not a function of E_{kin} alone: for the S star, $m_{\min}^A/E_{\text{kin}} = 0.0735, 0.0753$ and 0.0777 g/J at $v_0 = 10, 15$ and 20 m/s (D star: $0.0898, 0.0903, 0.0914$ g/J; star with ring at $n_s = 10$: $0.147\text{--}0.153$ g/J for S and $0.182\text{--}0.189$ g/J for D), a weak speed dependence of 1.8–6% over the range (Fig. 7a). The worst-case utilizations at $v_0 = 15$ m/s are $\eta_A = 0.128$ (S) and 0.192 (D) for the star, between the quasi-static limits of Sec. IV B (drone held and drone free in the plane) for S and slightly above the free limit for D (S: 0.075 and 0.149 ; D: 0.078 and 0.186); for the star with ring they are 0.064 (S) and 0.095 (D). For criterion A the worst case is $a = R/2$ for both nets and materials. *Star versus ring.* Under criterion A the ring does not pay for its own mass: at equal resolution ($n_s = 10$) the star with ring needs 1.9 (S) to 2.0 (D) times the mass of the plain star.

Sacrificial failure. For the S star, allowing one sacrificial break halves the minimum mass: $m_{\min}^A/m_{\min}^{\text{Bany}} = 2.04\text{--}2.10$, and the saving survives the localization requirement with $R_{\max} = R/2$ ($2.02\text{--}2.08$). For the D star the saving is $1.01\text{--}1.05$ under both criteria. In every case exactly one thread fails at $m_{\min}^B$ during the capture stroke, the impacted

TABLE XIII. Test 4: convergence for the off-centre impact of the reference case (star, $N = 8$, material S, $\epsilon_p = 0.1\epsilon_b$, $M = 1$ kg, $v_0 = 15$ m/s, $a = R/2$, gripped contact). Δ is the relative difference with respect to the finest discretization; η is evaluated at arrest, Eq. (31). No segment fails in this case. Net mass $m_{\text{net}} = 10.4$ g; at arrest ($n_s = 40$) the elastic energy is 80.7 J, the drone kinetic energy 30.8 J (in-plane motion), the net kinetic energy 0.72 J and the contact energy 0.27 J, out of $E_{\text{kin}} = 112.5$ J; the energy error is 0.02 J.

n_s (segments/thread)	w_{max}/R	Δw_{max} (%)	broken segments	η	$\Delta\eta$ (%)	CPU time (s)
10	0.4679	1.39	0	0.0757	1.28	0.37
20	0.4632	0.36	0	0.0749	0.24	0.75
40	0.4619	0.09	0	0.0747	0.01	1.89
80	0.4615	—	0	0.0747	—	5.97

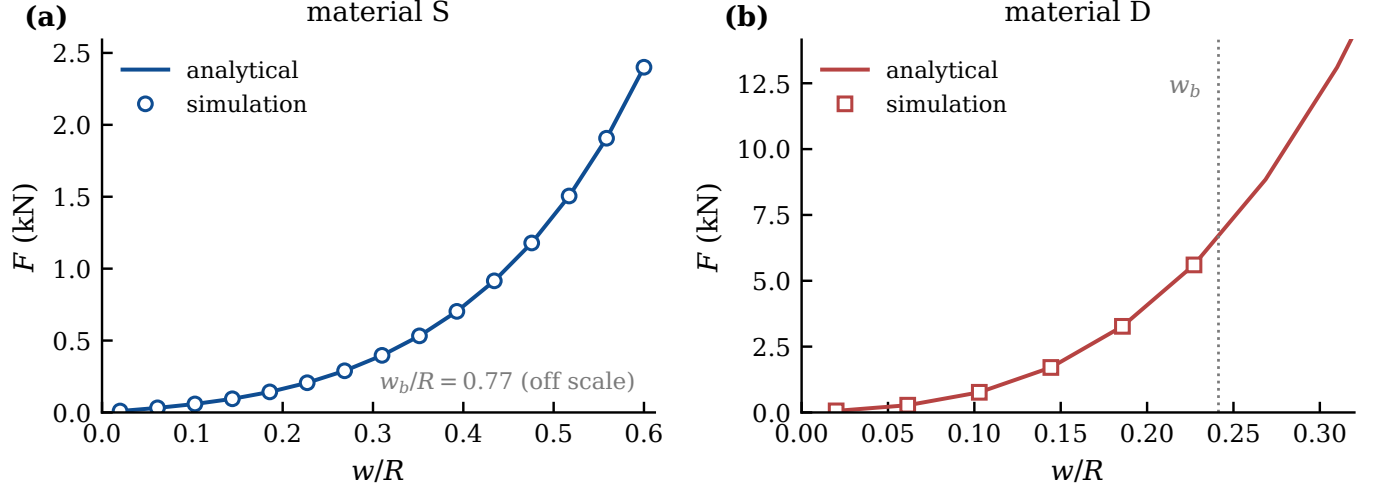


FIG. 4. Restoring force versus hub deflection for a star net with $N = 8$ radials, $R = 1$ m, $\epsilon_p = 0.1\epsilon_b$: (a) material S, (b) material D. Lines: Eq. (12); markers: quasi-static simulation. Dotted line: failure at w_b (Table III); for S, $w_b/R = 0.77$ lies beyond the plotted range.

radial (Fig. 6); for S the first break is the segment next to the drone, for D the segment at the frame. The worst case for B_{any} moves to $a = R/4$ for S, while it stays at $a = R/2$ for D. With the ring, the saving is 1.55–1.66 for S and 1.08–1.11 for D. The quasi-static estimate of Eq. (29) predicts almost the same saving for both materials with the drone held (3.3 for S, 3.0 for D), and 2.0 (S) and 3.2 (D) with the drone free in the plane; the S saving equals the free-limit value, while for D the simulations give a third of it. With the stricter $R_{\text{max}} = R/4$, the S saving is 1.03 at $v_0 = 10$ and 20 m/s but 1.95 at 15 m/s: the first break lies close to $0.25R$ from the drone, so this value is a threshold effect and is not used. **[TO DO: Rareş and Antonia compare the dynamic saving with their quasi-static estimate of Eq. (29) for S and D, and explain the difference: compare η_A and η_B of the arrested runs with Fig. 5, and use the energy form of Λ_E of Sec. V C.]**

Why the linear material does not save mass. The energy records of the star at $m_{\text{min}}^{\text{Bany}}$ ($M = 1$ kg, $v_0 = 20$ m/s, $E_{\text{kin}} = 200$ J, $a = R/2$, $n_s = 40$) show two different sequences. In S the impacted radial breaks at 19.9 ms, when the drone still carries 142 J and the net stores 62 J; 5 ms later the drone has 132 J, almost all of it as normal velocity (16.1 m/s), and the intact net absorbs it. In D the radial breaks at 10.7 ms, when the drone carries only 7.7 J and practically all of E_{kin} is stored in the net; 5 ms later the drone has 136 J again, now as in-plane velocity of 16.5 m/s towards the hub (normal velocity -0.6 m/s). The inner segment and the hub act as a sling that returns the stored energy to the drone, and the damaged net must absorb it a second time. Constraining the drone to move along z removes the sling: D then breaks at 6.3 ms with 124 J still in the drone, as S does, and the saving at the worst point becomes 3.18 for D and 3.46 for S, close to the quasi-static values with the drone held (3.0 and 3.3). The freedom to move in the plane lowers m_{min}^A by a factor 1.8 for S (27.45 to 15.53 g) and 2.6 for D (46.74 to 18.29 g), but changes $m_{\text{min}}^{\text{Bany}}$ little (S: 7.94 to 7.39 g) or raises it (D: 14.72 to 17.65 g). The material effect on the saving is therefore dynamic and lateral: the stiff linear net stores almost all of E_{kin} before its first break and releases it into the drone.

Impacts near the frame. Table XIV covers the central region $a \leq R/2$. Figure 8 extends the impact points to $a = 0.8R$. For S, m_{min}^A grows steeply towards the frame: at $v_0 = 20$ m/s from 15.5 g at $R/2$ to 46.5 g at $0.8R$ for

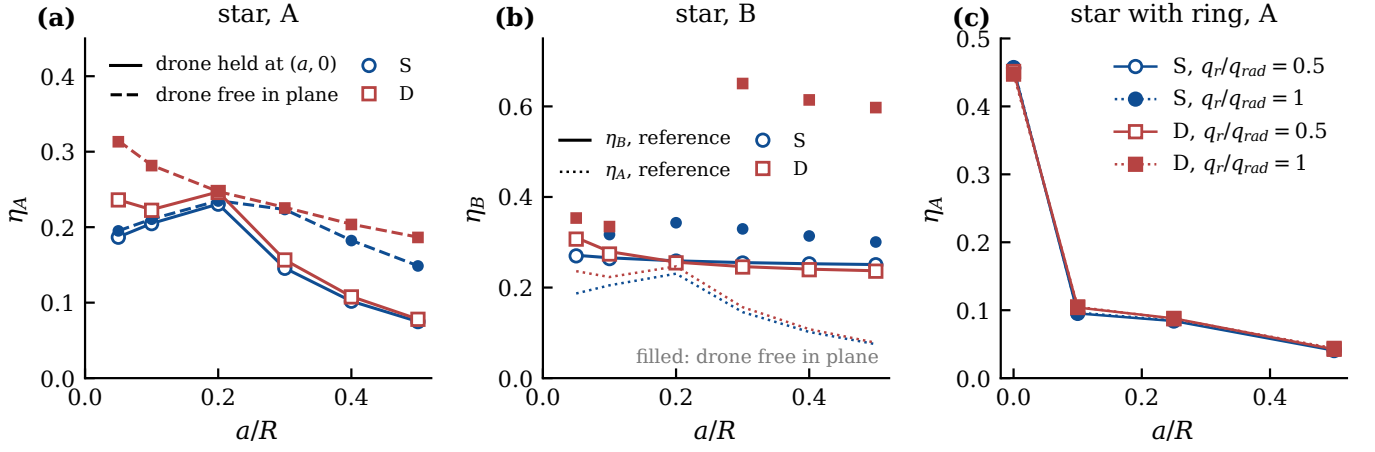


FIG. 5. Quasi-static material utilization versus impact position a/R along a radial, $N = 8$, $\epsilon_p = 0.1\epsilon_b$; circles: material S, squares: material D. (a) Star net, criterion A (first failure). Solid lines and open markers: gripped point held at $(a, 0)$; dashed lines and filled markers: gripped point free in the plane (no lateral force on the drone). (b) Star net, criterion B (one sacrificial outer segment), gripped point held (open markers) and free in the plane after the break (filled markers, simulator only); dotted lines: η_A of panel (a) for comparison. In (a,b), lines: reference minimization of Sec. IV B; markers: simulator. (c) Star with one ring at $R/2$, criterion A, gripped point held, for two ring-to-radial force-density ratios q_r/q_{rad} (simulator). The minimum over a enters Eq. (29).

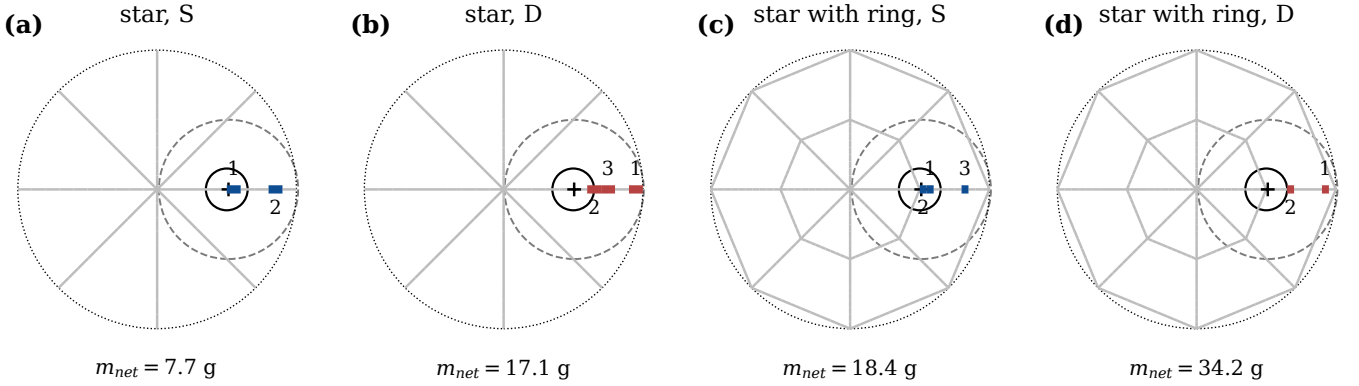


FIG. 6. Localization of failure: broken segments (thick, numbered in the order of failure) after arrest, for the star (a,b) and the star with one ring at $R/2$ (c,d), materials S (a,c) and D (b,d). Same drone ($M = 1$ kg, $v_0 = 20$ m/s, $r_d = 0.15$ m, solid circle at the impact point $a = R/2$), $n_s = 10$, net mass at the B_{any} minimum of this resolution (value below each panel). Cross: drone in-plane position at the first failure; dashed circle: $R_{max} = R/2$ around it; dotted circle: frame. In all four cases only the impacted radial fails, outside the drone.

the star (a factor 3.0, and 3.1 at 10 and 15 m/s) and from 30.6 to 97.3 g with the ring (3.2–3.3). The ring at $R/2$ does not protect the outer annulus. For D the growth is only 11–12% (star) and 21–25% (ring). The B masses do not grow: for the S star at 20 m/s, $m_{min}^{B_{any}} = 6.4, 6.0$ and 4.3 g at $0.6, 0.7$ and $0.8R$. If the design region extends to $0.8R$, the worst-case m_{min}^A of the S star triples and the saving $m_{min}^A/m_{min}^{B_{any}}$ rises to about 6 (46.5/7.5); for D it stays at 1.1. Near the frame the B values are less reliable. Of the 48 B values at $a = 0.8R$, 20 are discarded because the drone touches the net within one segment length of the frame before the reversal, and the value of the D star at 10 m/s (criteria B_{any} and B_{loc} with $R_{max} = 0.75R$, 0.44 g) because it lies below the energy bound $E_{kin}/e_{mat} = 0.87$ g: there the drone keeps 80% of E_{kin} as in-plane velocity when its normal velocity reverses, so the reversal is not a capture. Values whose threshold run keeps more than half of E_{kin} in the drone at the reversal are marked in Fig. 8; none of them sets a worst case for $a \leq R/2$. A net that is light in the centre and reinforced near the frame is the obvious design consequence, and the motivation for the topology search of Sec. VII.

Localization map. Figure 9 classifies 182 runs on the star ($v_0 = 20$ m/s, $a/R = 0.1$ – 0.7 , $m_{net}/m^* = 1.5$ – 10). For S a band of arrests with failure confined within $R/2$ of the drone separates the intact runs from the perforated ones over the whole range of a ; only 4 runs fail beyond $R/2$. For D this band is nearly absent at $a \geq 0.3R$: below the onset of

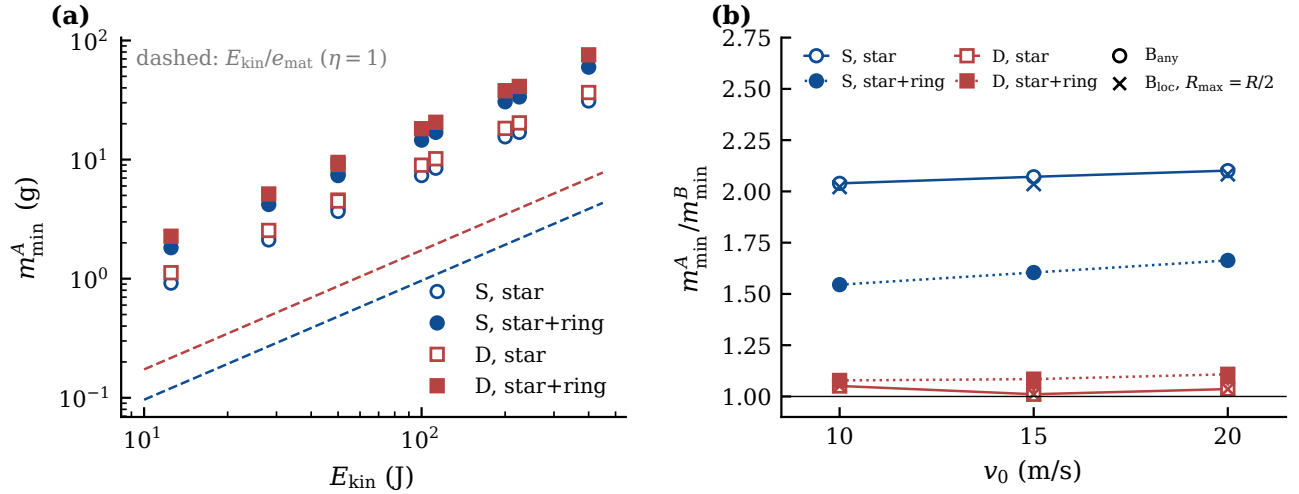


FIG. 7. (a) Minimum net mass under criterion A versus drone kinetic energy, worst case over the impact points, for the star (open) and the star with one ring at $R/2$ (filled), materials S (circles) and D (squares); $N = 8$, $R = 1$ m, $\epsilon_p = 0.1\epsilon_b$, $M \in \{0.25, 1, 2\}$ kg, $v_0 \in \{10, 15, 20\}$ m/s. Dashed lines: absolute lower bound $E_{\text{kin}}/e_{\text{mat}}$ ($\eta = 1$). (b) Mass saving due to sacrificial failure, $m_{\min}^A/m_{\min}^B$, versus impact speed for $M = 1$ kg: circles and squares, criterion B_{any} ; crosses, criterion B_{loc} with $R_{\text{max}} = R/2$. Star: $n_s = 40$; star with ring: $n_s = 10$.

TABLE XIV. Minimum net mass (grams) from Algorithm 2, worst case over the impact points $a/R \in \{0, 0.25, 0.5\}$ on radial 0, for the star and the star with one ring at $R/2$, and mass saving due to sacrificial failure under the two B criteria. $N = 8$ radials; $\epsilon_p = 0.1\epsilon_b$; $R = 1$ m; $M = 1$ kg; gripped contact; B_{loc} with $R_{\text{max}} = R/2$; $k_{\text{max}} = 10$ threads; bisection tolerance 1%. At $m_{\min}^B$, exactly one thread fails at the worst-case point of criterion A ($a = R/2$) in every row. For $M = 0.25$ and 2 kg, $m_{\min} \propto M$ at fixed v_0 within the bisection tolerance (checked at $n_s = 10$ for all rows and at $n_s = 40$ for $M = 0.25$ kg, $v_0 = 20$ m/s); the star row at $v_0 = 20$ m/s is obtained in this way from $M = 0.25$ kg; computed directly at $M = 1$ kg it gives the same $m_{\min}^A$, and $m_{\min}^{B_{\text{any}}}$ and $m_{\min}^{B_{\text{loc}}}$ 2% and 9% higher (Sec. VI). All criteria refer to the capture stroke; for impact points closer to the frame see Fig. 8.

Net	Material	n_s	v_0 (m/s)	$E_{\text{kin}}/e_{\text{mat}}$	$m_{\min}^A$	$m_{\min}^{B_{\text{loc}}}$	$m_{\min}^{B_{\text{any}}}$	$\frac{m_{\min}^A}{m_{\min}^{B_{\text{loc}}}}$	$\frac{m_{\min}^A}{m_{\min}^{B_{\text{any}}}}$
star	S	40	10	0.48	3.67	1.82	1.80	2.02	2.04
star	S	40	15	1.08	8.47	4.16	4.09	2.04	2.07
star	S	40	20	1.93	15.53	7.46	7.39	2.08	2.10
star	D	40	10	0.87	4.49	4.27	4.27	1.05	1.05
star	D	40	15	1.95	10.16	10.06	10.06	1.01	1.01
star	D	40	20	3.46	18.29	17.65	17.65	1.04	1.04
star+ring	S	10	10	0.48	7.35	4.75	4.75	1.55	1.55
star+ring	S	10	15	1.08	16.93	10.55	10.55	1.60	1.60
star+ring	S	10	20	1.93	30.58	18.39	18.39	1.66	1.66
star+ring	D	10	10	0.87	9.09	8.43	8.43	1.08	1.08
star+ring	D	10	15	1.95	20.57	19.60	18.97	1.05	1.08
star+ring	D	10	20	3.46	37.87	34.85	34.17	1.09	1.11

failure the runs perforate or time out, consistent with $m_{\min}^B \approx m_{\min}^A$. The exceptions, arrested runs of D at $a \geq 0.5R$ and $m_{\text{net}} \leq 2.5m^*$, below the minimum masses, are not captures: after the first break the drone keeps 78–87% of E_{kin} and is thrown sideways, and its normal velocity reverses during this motion.

Frictionless sliding. With the thread sliding freely over the drone (star, $n_s = 40$, $v_0 = 20$ m/s), the saving disappears: $m_{\min}^A = 21.19$ g and $m_{\min}^{B_{\text{any}}} = 21.40$ g for S, 18.61 and 18.44 g for D. The impacted radial then carries the same tension on both sides of the drone and fails as a whole, so there is no short sacrificial segment. These values are indicative: with sliding contact η at arrest changes by 12% between $n_s = 40$ and 80 (Table XIII).

After the capture stroke. To test the arrest rule, every threshold run (the run at $m_{\min}^X$ for each row, impact point

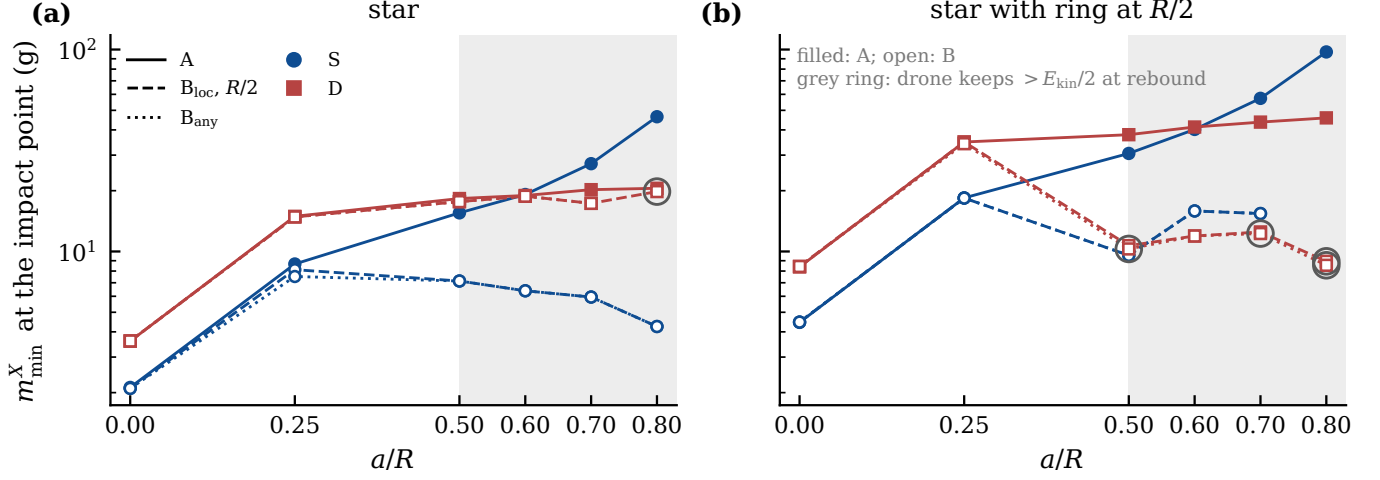


FIG. 8. Minimum net mass at each impact point a/R on radial 0, from the central region (Table XIV, $a \leq R/2$) to the shaded annulus near the frame, for the star ($n_s = 40$) and the star with one ring at $R/2$ ($n_s = 10$); $M = 1$ kg, $v_0 = 20$ m/s, $N = 8$, $\epsilon_p = 0.1\epsilon_b$. Solid lines and filled markers: criterion A; dashed: B_{loc} with $R_{\text{max}} = R/2$; dotted: B_{any} . Values discarded because the drone touches the net next to the frame before the reversal are omitted (B for the ring with S at $0.8R$). Grey rings: threshold runs in which the drone keeps more than half of E_{kin} at the reversal of its normal velocity. All values refer to the capture stroke.

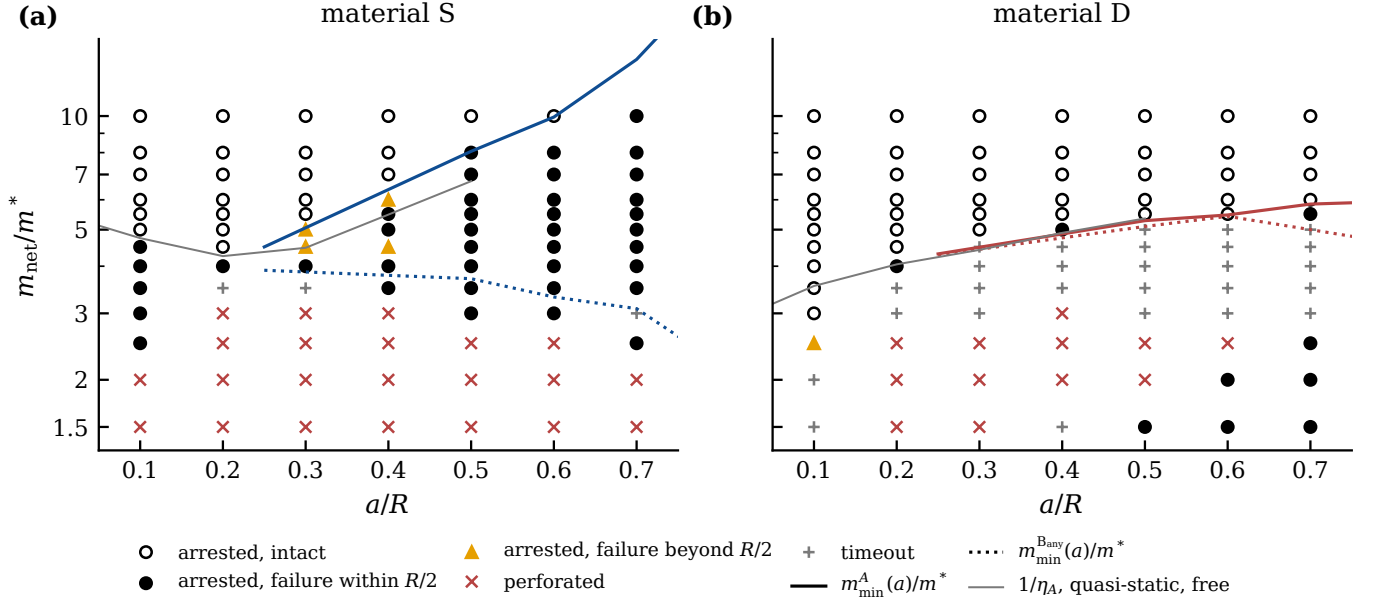


FIG. 9. Localization map for the off-centre impact on the star, (a) material S and (b) material D: outcome of single runs versus impact position and net mass in units of the absolute bound $m^* = E_{\text{kin}}/e_{\text{mat}}$; $M = 1$ kg, $v_0 = 20$ m/s, $n_s = 40$. Coloured lines: $m_{\text{min}}^A(a)$ (solid) and $m_{\text{min}}^{B_{\text{any}}}(a)$ (dotted) from Fig. 8. Grey line: quasi-static $1/\eta_A$ with the drone free in the plane (Fig. 5a). Arrest is the reversal of the normal velocity of the drone (capture stroke).

and criterion) was continued to t_{end} after the first reversal. At the reversal the drone keeps a median 3–5% of E_{kin} for $a \leq R/2$, so the capture itself is genuine. In the undamped model, however, the net returns its elastic energy and the drone oscillates, and at the minimum mass there is no margin: for $a \leq R/2$ at least one further thread breaks in 33 of 36 runs for criterion A and in 30–33 of 36 for the B criteria (median 1 further thread for A and 2–3 for B), and in 62 of the 180 runs at least eight threads break by t_{end} . No drone perforates within t_{end} for $a \leq R/2$. The minimum masses of Table XIV are therefore minimum *capture* masses. Retaining the drone requires dissipation, or a margin on the mass, that this model does not quantify (Sec. IX).

TABLE XV. Surrogate and descriptor performance on the test splits of Table XI. Baselines: ?.

Model	Split	$R^2(\eta)$	Arrest acc. (%)	Edge-failure F1	$\Delta R_d/R_d$ (%)	Speed-up
Spectral descriptors	interp.	?	?	—	—	?
Spectral descriptors	unseen topol.	?	?	—	—	?
GNN	interp.	?	?	?	?	?
GNN	unseen topol.	?	?	?	?	?

Reference solution (Tavi only). *Analysis by Tavi, from the energy records of the star runs at $m_{\min}^{\text{Bany}}$ ($M = 1$ kg, $v_0 = 20$ m/s, $E_{\text{kin}} = 200$ J, $a = R/2$, $n_s = 40$; the $n_s = 10$ runs of Fig. 6a,b give the same picture). Not yet a derivation.*

	S	D
time of first failure (ms)	19.9	10.7
elastic energy of the net just before it, incl. prestress (J)	62	202
drone kinetic energy just before it (J)	142	7.7
drone kinetic energy 5 ms after it (J)	132	136
in-plane / normal drone speed 5 ms after it (m/s)	1.7 / 16.1	16.5 / 0.6
failure energy of the broken segments (J)	9.6	9.6
kinetic energy of the threads 5 ms after it (J)	36	55

S. The outer segment reaches ϵ_b early, when the net stores only about 30% of E_{kin} : the stiffening law concentrates the strain in the short outer segment while the rest of the net is still soft. After the break the drone keeps moving downwards at almost the same speed, and the intact net absorbs the remaining energy gradually; the drone drifts slowly towards the hub (from $x = 0.51R$ to $0.27R$ at arrest). The broken segment works as intended: it absorbs its share and then leaves the rest of the net to finish the arrest. *D.* The outer segment fails only when the whole net stores practically all of E_{kin} : the linear stiff material spreads the strain over the hub and all radials. At the break, most of this elastic energy is returned to the drone, not dissipated: its kinetic energy rises from 7.7 to 136 J within 5 ms, almost entirely as in-plane velocity towards the hub (16.5 m/s, while $v_z \approx -0.6$ m/s). The inner segment and the hub act as a sling. The damaged net then has to absorb almost two thirds of E_{kin} a second time, now near the hub, so the sacrificial break adds little capacity and $m_{\min}^{\text{B}} \approx m_{\min}^{\text{A}}$. *Consequence for the analysis.* The quasi-static estimate of Eq. (29) assumes that the energy stored in the intact net at the first failure stays in the net. This holds for S, where little is stored, and fails for D, where most of it is released into the drone. A corrected estimate must (i) compute the energy stored in the intact net at the first failure, $U_{\text{int}}(w_b)$; (ii) treat the release as an elastic rebound that returns U_{int} to the drone, minus the failure energy and the energy radiated into the threads; and (iii) evaluate the capacity of the damaged net for a drone arriving near the hub with in-plane velocity. The runs with the drone constrained to move along z confirm (ii): without the in-plane degree of freedom there is no sling, D breaks early as S does, and the saving returns to the quasi-static value with the drone held (3.18 against 3.0). Point (iii) is still open; the mass scan around $m_{\min}^{\text{B}}$ ($\mu = 0.9\text{--}1.5$) shows the same sequence at all masses where D fails.

C. Spectral descriptors and learned surrogate

[TO DO: Octavian: results text.]

IX. DISCUSSION

Tavi's note: Written in advance as a frame. Statements marked with ? depend on the results and must be confirmed or corrected. Rareş and Antonia rewrite it in their own words.

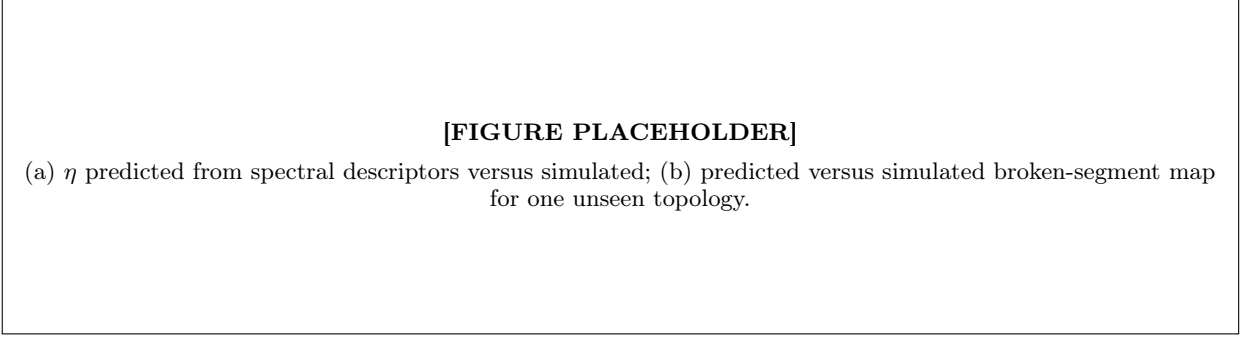


FIG. 10. (a) Material utilization predicted from the first $k = ?$ generalized eigenvalues of Eq. (5) versus simulation. (b) Per-edge failure predicted by the GNN surrogate versus Algorithm 1, for an unseen topology.

A. What this work adds

Compared with Cranford *et al.* [1], whose analytical argument is quasi-static and continuum-based, the present work (i) includes the dynamic load transfer at a sudden break; (ii) uses the cubic law of Eq. (2), which is not a power law; (iii) treats a discrete anchored net and quantifies the material saving due to sacrificial failure. Compared with the dynamic amplification factors of composites [10] and bridges [11], computed for linear elements, we show that stiffening reduces the *strain* overshoot. The regime analysis of Sec. IIID shows that for drone impacts the classical wave theory [3] sets only the local, fast events, and that the minimum mass for hub impacts is independent of the number and length of radials. Sacrificial failure halves the minimum capture mass of a star net made of the stiffening material ($m_{\min}^A/m_{\min}^B = 2.0\text{--}2.1$ for impacts within $R/2$, also when the damage must stay within $R/2$ of the drone), while for the linear material the saving is at most 5%, because the stiff net stores almost all of the kinetic energy before its first break and returns it to the drone as in-plane velocity. Without failure, the mass needed for impacts near the frame is three times that for the central region with the stiffening material, so the outer annulus, not the centre, sets the design. **[TO DO: State whether the criterion $\Lambda_{\text{loc}} = 1$ matches the simulations within ?%.]**

B. What the spectral descriptors and the surrogate add

[TO DO: Octavian: do spectral descriptors predict η and localization? Where do they fail (isospectral nets, strongly nonlinear regime)? Speed-up for design sweeps and inverse design.]

C. Potential applications

- **Counter-drone barriers.** Fixed nets around sensitive sites, where the design target is the lightest (or least visible) barrier for a given threat [20].
- **Other arresting nets.** Rockfall barriers, safety nets, and aircraft arresting systems, where localized failure could replace heavier dissipators.
- **Bio-inspired design rules.** When a stiffening fiber pays off, and when a linear high-strength fiber is better, depending on whether single-use or multi-hit operation is required.

D. Limitations

- The cubic law has no yield region; real dragline silk softens before stiffening [1].
- Threads are massless in the quasi-static analysis, and bending, friction at crossings, and knots are ignored.
- There is no dissipation apart from thread failure and the capture of the gripped node. The minimum masses are capture masses: at these masses a further thread usually breaks after the first rebound (Sec. VIII). Retention of the drone depends on material hysteresis, aerodynamic drag, rotor thrust, and gravity, none of which is modeled.

- The drone is a rigid sphere and entanglement is idealized as a single gripped point; propellers and arms are not modeled. The opposite limit, frictionless sliding of the thread over the drone, removes the sacrificial saving in a pure star (Sec. VIII); contact with friction lies between the two limits.
- For impacts at $0.8R$ the B criteria are sensitive to contact between the drone and the threads next to the frame, and part of these values is discarded; the A values near the frame are not affected.
- The exact finite-amplitude junction solution is confirmed by the net simulation only up to incident amplitudes of $0.3\epsilon_b$ and, for the break threshold, for material S with $N \geq 8$.
- Only perpendicular impacts are treated, and only two net families are studied analytically.
- Strength variability between threads, which drives cascades in fiber-bundle models [13], is not included.
- Failure is instantaneous at ϵ_b , without damage or softening. A broken radial then fails at several segments as the released thread recoils; the arrest criteria count threads and are not affected, but the energy dissipated by failure and the shape of the failure maps depend on this idealization.
- The star with ring is computed with $n_s = 10$; at this resolution the star masses are up to 7% higher than at $n_s = 40$, which does not change the ring-to-star mass ratio of about two. For the ring itself, $n_s = 20$ changes the masses by at most 1.6%.

E. Future work

1. Retention after capture: a hysteretic constitutive law, gravity, and an arrest criterion over the whole oscillation, giving a minimum retention mass and the dissipation per cycle needed for the first stroke to remain the critical one.
2. Reinforcement of the outer annulus, where the minimum mass without failure is set.
3. Multiple-impact survival (swarms): the residual capacity of a net after sacrificial failure.
4. Oblique impacts, and drones modeled as deformable bodies with rotors.
5. Statistical thread strength and the transition to fiber-bundle behavior.
6. Laboratory drop tests on small nets of fishing line or Dyneema line.
7. Inverse design of topologies and section distributions with the learned surrogate.

X. CONCLUSIONS

We studied the arrest of small drones by anchored, prestressed nets and asked how much material sacrificial failure can save. Generalizing the Smith theory of transverse impact to a strain-stiffening cubic law, we showed that drone impacts are quasi-static: the initial waves carry less than 1% of the failure strain, and transverse waves cross the net tens of times during arrest. For hub impacts on radial nets, the minimum mass is $E_{\text{kin}}/(\eta_A e_{\text{mat}})$ with $\eta_A = 1 - \Phi(\epsilon_p)/\Phi(\epsilon_b)$, independent of the number and length of radials. We derived the transmission of tension waves at junctions and the dynamic overload after a sudden break, which is smaller in strain for stiffening responses, and combined them into a localization number Λ_{loc} . Simulations reproduce the analytical wave speeds and strains within 0.3% below 500 m/s and the quasi-static utilizations within 1.5%. For impacts in the central half of the net, allowing localized failure reduces the minimum capture mass of a star net by a factor 2.0–2.1 for the silk-like material and by at most 5% for the linear one; the linear net loses its saving because the energy stored before the first break is released into a lateral sling of the drone. Near the frame the mass required without failure is three times larger for the silk-like material. In the undamped model the net at the minimum capture mass usually loses further threads on the rebound, so retention requires dissipation, which we leave to future work. Spectral descriptors of the prestressed net predict the material utilization with $R^2 = ?$, and a graph-neural-network surrogate predicts thread failure on unseen topologies with F1 = ?, at a cost ? times lower than the full simulation.

ACKNOWLEDGMENTS

[TO DO: Funding, computing resources, and people who helped.]

AUTHOR CONTRIBUTIONS

[TO DO: CRediT taxonomy, e.g.: R.M. and A.R.: formal analysis, validation, writing (Secs. II–V, VIII A–B). O.-M.R.: conceptualization, supervision, software (simulator and surrogate), writing (Secs. VI–VII, VIII C), review and editing.]

DATA AVAILABILITY

[TO DO: Repository (e.g. Zenodo, GitHub) with simulator code, HDF5 data set, and trained models; DOI.]

Appendix A: Derivation of the Smith relation for a general law

[TO DO: Rares and Antonia: full derivation of Eqs. (7)–(9) from mass and momentum balance across both fronts, including the proof that the strain is continuous across the kink, and the limits $\epsilon \ll 1$ (linear law) and $\epsilon \gg 1$ (cubic law).]

Reference solution (Tavi only). *Jump conditions.* Use the Lagrangian coordinate X and the position $\mathbf{x}(X, t)$; strain $1 + \epsilon = |\partial \mathbf{x} / \partial X|$, unit tangent $\hat{\mathbf{t}} = (\partial \mathbf{x} / \partial X) / (1 + \epsilon)$, velocity $\mathbf{v} = \partial \mathbf{x} / \partial t$, nominal tension $T(\epsilon)$ along $\hat{\mathbf{t}}$, linear density μ_0 in the rest state. For a front moving at speed c in X , with $[f] = f_{\text{behind}} - f_{\text{ahead}}$:

$$\text{continuity of } \mathbf{x} : [\mathbf{v}] = -c[(1 + \epsilon)\hat{\mathbf{t}}], \quad \text{momentum: } [T\hat{\mathbf{t}}] = -\mu_0 c[\mathbf{v}].$$

Longitudinal front ($X = c_L t$): ahead $\epsilon = 0$, $\mathbf{v} = 0$; behind ϵ , $\mathbf{v} = (-u_2, 0)$, $\hat{\mathbf{t}} = (1, 0)$. Continuity: $u_2 = c_L \epsilon$. Momentum: $T(\epsilon) = \mu_0 c_L u_2$. Hence Eq. (7), $c_L^2 = T(\epsilon) / (\mu_0 \epsilon)$ (a chord speed; it reduces to $\sqrt{E_0 / \rho}$ for the linear law). *Transverse front* ($X = c_T t$): eliminating $[\mathbf{v}]$ gives $[T\hat{\mathbf{t}}] = \mu_0 c_T^2 [(1 + \epsilon_1)\hat{\mathbf{t}}_1 - (1 + \epsilon_2)\hat{\mathbf{t}}_2]$, i.e. $T_1 \hat{\mathbf{t}}_1 - T_2 \hat{\mathbf{t}}_2 = \mu_0 c_T^2 [(1 + \epsilon_1)\hat{\mathbf{t}}_1 - (1 + \epsilon_2)\hat{\mathbf{t}}_2]$. Since $\hat{\mathbf{t}}_1$ and $\hat{\mathbf{t}}_2$ are not parallel, the coefficients must match: $T(\epsilon_1) / (1 + \epsilon_1) = T(\epsilon_2) / (1 + \epsilon_2) = \mu_0 c_T^2$. *Continuity of strain.* $g(\epsilon) = T(\epsilon) / (1 + \epsilon)$ is strictly increasing for the cubic law: $g'(1 + \epsilon)^2 = T'(1 + \epsilon) - T = E_0 + 3b\epsilon^2 + 2b\epsilon^3 > 0$. Hence $\epsilon_1 = \epsilon_2 = \epsilon$ and $c_T^2 = T(\epsilon) / [\mu_0(1 + \epsilon)]$, Eq. (8). (The same holds for any law with $T'(1 + \epsilon) > T$, which includes all convex laws with $T(0) = 0$.) *Kinematics.* Region 1 is straight and uniformly strained, so all its particles move with the impact point, $\mathbf{v}_1 = (0, v_0)$. Let $\hat{\mathbf{t}}_1 = (\cos \theta, -\sin \theta)$. The continuity condition at the kink gives $(u_2, v_0) = c_T(1 + \epsilon)(1 - \cos \theta, \sin \theta)$, i.e. $\cos \theta = 1 - c_L \epsilon / [c_T(1 + \epsilon)]$ and $\sin \theta = v_0 / [c_T(1 + \epsilon)]$. Then $\sin^2 \theta + \cos^2 \theta = 1$ gives the first form of Eq. (9), which is the Pythagorean relation of Sec. III obtained here without a geometric argument. Using $c_T^2(1 + \epsilon)^2 = T(1 + \epsilon) / \mu_0$ and $c_L \epsilon = \sqrt{T \epsilon / \mu_0}$: $v_0^2 = 2c_T(1 + \epsilon)c_L \epsilon - c_L^2 \epsilon^2 = (T / \mu_0)[2\sqrt{\epsilon(1 + \epsilon)} - \epsilon]$. *Limits.* $\epsilon \ll 1$, $T = E_0 A \epsilon$: $v_0^2 \simeq 2c^2 \epsilon^{3/2}$, $\epsilon \simeq (v_0^2 / 2c^2)^{2/3}$, and the exact linear form is Smith's $v_0 = c\sqrt{2\epsilon\sqrt{\epsilon(1 + \epsilon)} - \epsilon^2}$. $\epsilon \gg 1$, $T \simeq \beta \epsilon^3$: $2\sqrt{\epsilon(1 + \epsilon)} - \epsilon \simeq \epsilon + 1 \simeq \epsilon$, so $v_0^2 \simeq \beta \epsilon^4 / \mu_0$, $\epsilon \propto v_0^{1/2}$, $T \propto v_0^{3/2}$. This limit is formal ($\epsilon_b = 0.3$); for S at failure the linear and cubic terms are comparable ($b\epsilon_b^2 / E_0 = 0.99$), so the full relation must be used.

Appendix B: Hub impact on a star net: derivation of Eqs. (11)–(13)

a. Assumptions. (i) The net plane is vertical and the drone flies horizontally, along the normal $\hat{\mathbf{z}}$ to the net, so gravity and rotor thrust do not work along the motion and are neglected. (ii) The drone touches the net only at the hub (point contact, $r_d \ll R$). (iii) The response is quasi-static: thread inertia is neglected, which is justified by $\Pi_E \sim 10^{-3}$ (Sec. IIID); the radials are therefore straight at all times. (iv) The $N \geq 2$ radials are identical and equally spaced in angle, $\varphi_k = 2\pi k / N$, and the frame is rigid. (v) $T(\epsilon)$ is the nominal tension, i.e. the force per thread expressed in terms of the rest cross-section, so that $T = A\sigma$ and the strain energy per unit rest volume is $\Phi(\epsilon)$ of Eq. (3).

b. Geometry and strain [Eq. (11)]. Before impact the hub is at the centre of the frame and each radial spans the distance R at the prestrain ϵ_p , so its rest length is $\ell = R/(1 + \epsilon_p)$. When the hub is displaced by w along $\hat{z}$, the anchor of radial k stays at $R(\cos \varphi_k, \sin \varphi_k, 0)$ and the hub is at $(0, 0, w)$. The current length of every radial is $L(w) = \sqrt{R^2 + w^2}$, independent of k , and the strain is

$$\epsilon(w) = \frac{L(w)}{\ell} - 1 = (1 + \epsilon_p) \sqrt{1 + (w/R)^2} - 1, \quad (\text{B1})$$

which is Eq. (11) and is exact under assumptions (ii)–(iv). Expanding the square root for $w \ll R$, $(1 + \epsilon_p)(1 + w^2/2R^2) = 1 + \epsilon_p + w^2/2R^2 + \epsilon_p w^2/2R^2$; dropping the last term, of relative order ϵ_p , gives the approximate form $\epsilon \simeq \epsilon_p + w^2/2R^2$. For material S at failure $w_b/R \approx 0.65\text{--}0.83$ (Table III), so only the exact form is used in this work.

c. Restoring force [Eq. (12), left]. Radial k pulls the hub towards its anchor with the force $T(\epsilon) \hat{t}_k$, where

$$\hat{t}_k = \frac{R(\cos \varphi_k, \sin \varphi_k, 0) - (0, 0, w)}{\sqrt{R^2 + w^2}}. \quad (\text{B2})$$

All radials carry the same tension because they have the same strain. The in-plane components cancel, since $\sum_{k=0}^{N-1} (\cos \varphi_k, \sin \varphi_k) = 0$ for equally spaced radials with $N \geq 2$. The out-of-plane components add up to the force that opposes the motion,

$$F(w) = N T(\epsilon(w)) \sin \theta = N T(\epsilon(w)) \frac{w}{\sqrt{R^2 + w^2}}, \quad \tan \theta = \frac{w}{R}. \quad (\text{B3})$$

This is the left part of Eq. (12).

d. Stored energy [Eq. (12), right]. The work done on the net from $w = 0$ to w is $U(w) = \int_0^w F(w') dw'$. Differentiating the strain,

$$\frac{d\epsilon}{dw} = (1 + \epsilon_p) \frac{w/R^2}{\sqrt{1 + (w/R)^2}} = \frac{(1 + \epsilon_p) w}{R \sqrt{R^2 + w^2}} = \frac{1}{\ell} \frac{w}{\sqrt{R^2 + w^2}}, \quad (\text{B4})$$

where $\ell(1 + \epsilon_p) = R$ was used. Hence $F(w) = N \ell T(\epsilon) d\epsilon/dw = N \ell A \sigma(\epsilon) d\epsilon/dw$, and with $d\Phi/d\epsilon = \sigma$,

$$U(w) = N \ell A \int_{\epsilon_p}^{\epsilon(w)} \sigma(\epsilon') d\epsilon' = N \ell A [\Phi(\epsilon(w)) - \Phi(\epsilon_p)]. \quad (\text{B5})$$

This is the right part of Eq. (12). It is the strain energy added to the rest volume $N \ell A$ of the net, as expected for a conservative elastic system, and it holds for any law $T(\epsilon)$ and without the small-angle approximation.

e. Failure, utilization, and minimum mass [Eq. (13)]. All radials reach ϵ_b at the same deflection w_b , given by $\epsilon(w_b) = \epsilon_b$:

$$\frac{w_b}{R} = \sqrt{\left(\frac{1 + \epsilon_b}{1 + \epsilon_p}\right)^2 - 1}. \quad (\text{B6})$$

Under criterion A the drone is arrested if its kinetic energy is stored before failure, $E_{\text{kin}} \leq U(w_b) = N \ell A [\Phi(\epsilon_b) - \Phi(\epsilon_p)]$. Since $m_{\text{net}} = N \rho A \ell$,

$$\frac{U(w_b)}{m_{\text{net}}} = \frac{\Phi(\epsilon_b) - \Phi(\epsilon_p)}{\rho} = e_{\text{mat}} \left[1 - \frac{\Phi(\epsilon_p)}{\Phi(\epsilon_b)} \right] = e_{\text{mat}} \eta_A, \quad (\text{B7})$$

with $e_{\text{mat}} = \Phi(\epsilon_b)/\rho$ from Eq. (3). The arrest condition becomes $m_{\text{net}} \geq E_{\text{kin}}/(\eta_A e_{\text{mat}})$, which is Eq. (13). Because $U(w_b)$ and m_{net} are both proportional to the rest volume $N \ell A$, their ratio does not depend on N , R , or A separately: these parameters fix only the mesh size and the stopping distance w_b .

f. Checks. For $\epsilon_p = 0$, $\eta_A = 1$: every element of the net fails simultaneously at ϵ_b , so the material is fully used. With the parameters of Table I, $w_b/R = \sqrt{1.3^2 - 1} = 0.831$ for S and $\sqrt{1.032^2 - 1} = 0.255$ for D, as in Table III. For D ($b = 0$), $\eta_A = 1 - (\epsilon_p/\epsilon_b)^2$, which gives 0.990 and 0.910 for $\epsilon_p/\epsilon_b = 0.1$ and 0.3.

g. Remarks. The result is an upper bound on the performance of real star nets: a finite drone radius, off-centre contact, thread inertia at the moment of first contact, and the dynamic overshoot of Sec. VB all reduce the energy stored at failure. **[TO DO: Rareş and Antonia: give $w_{\max}(E_{\text{kin}})$ in closed form for D, and numerically for S.]**

Reference solution (Tavi only). Arrest at $w_{\max}$ requires $U(w_{\max}) = E_{\text{kin}}$. With $N\ell A = m_{\text{net}}/\rho$ and $m^* = E_{\text{kin}}/e_{\text{mat}}$,

$$\Phi(\epsilon_{\max}) = \Phi(\epsilon_p) + \Phi(\epsilon_b) \frac{m^*}{m_{\text{net}}}, \quad \frac{w_{\max}}{R} = \sqrt{\left(\frac{1 + \epsilon_{\max}}{1 + \epsilon_p}\right)^2 - 1}.$$

For D ($b = 0$): $\epsilon_{\max} = \sqrt{\epsilon_p^2 + \epsilon_b^2 m^*/m_{\text{net}}}$. For S the equation is quadratic in $x = \epsilon_{\max}^2$, $\frac{b}{4}x^2 + \frac{E_0}{2}x - C = 0$ with $C = \Phi(\epsilon_p) + \Phi(\epsilon_b)m^*/m_{\text{net}}$, so a closed form also exists: $\epsilon_{\max}^2 = (\sqrt{E_0^2 + 4bC} - E_0)/b$. The result depends only on m_{net}/m^* , not on N or R separately, and $w_{\max} \propto R$. The TODO is well posed only once m_{net} is fixed: for $m_{\text{net}} = m_{\text{min}}^A$, $\epsilon_{\max} = \epsilon_b$ by definition. Values for $\epsilon_p = 0.1\epsilon_b$ and $m_{\text{net}} = \mu m_{\text{min}}^A$:

μ	S		D	
	$\epsilon_{\max}/\epsilon_b$	$w_{\max}/R$	$\epsilon_{\max}/\epsilon_b$	$w_{\max}/R$
1.0	1.000	0.770	1.000	0.241
1.5	0.857	0.700	0.819	0.215
2.0	0.764	0.652	0.711	0.198

Doubling the mass reduces the stopping distance of S only by 15%, because of stiffening.

Appendix C: Dynamic overload for a power-law response

For $F = kw^n$ and a sudden constant load $P = kw_s^n$ from rest at $w = 0$, Eq. (26) reads $kw_m^{n+1}/(n+1) = kw_s^n w_m$, which gives Eq. (27). For the loss of one of N elements from equilibrium at w_0 , with $r = w_m/w_0$,

$$\frac{N-1}{N} \frac{r^{n+1} - 1}{n+1} = r - 1, \quad (\text{C1})$$

whose root $r > 1$ is reported in Table IX as $\epsilon_m/\epsilon_0 = r^2$ (star geometry, $\epsilon_p = 0$).

Appendix D: Simulation parameters

Appendix E: List of symbols

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TABLE XVI. Numerical parameters used in all simulations.

Parameter	Symbol	Value
Segments per thread	n_s	40 (star); 10 (star with ring, Table XIV); 20 and 40 as checks); 800 (Test 1)
CFL number	C	0.5
Contact mode	—	gripped (Sec. II)
Contact penalty stiffness (Hertz-type, force $\propto \delta^{3/2}$)	k_c	S: 10^7 N/m $^{3/2}$; D: relative, contact frequency at δ_{ref} equal to $f_c = 8$ times the axial frequency of a segment (Sec. VI)
Reference penetration	δ_{ref}	$0.01 r_d = 1.5$ mm
Viscous damping (quasi-static tests 2 and 5 only)	—	400 s^{-1} per unit mass; 0 in all impact runs
Frame radius	R	1 m
Drone radius	r_d	0.15 m
Localization radius and maximum failures (criterion B)	$R_{\text{max}}, k_{\text{max}}$	$0.25R, 0.5R, 0.75R$; 10 threads
Bisection tolerance	δ	0.01 (all criteria)
Impact points per topology	$ \{\mathbf{p}_k\} $	3: $a/R = 0, 0.25, 0.5$ on radial 0 (Table XIV); 6, adding 0.6, 0.7, 0.8 (Fig. 8)
Simulated time	t_{end}	0.25 s; 0.5 s for the timeout checks
Output interval	Δt_{out}	5 ms in the impact runs; 0.1 ms for the energy records
Code and version	—	netsim , commit e65b141 (Table XIV); a6c4705 , a351fff and c2d1123 (Figs. 8, 9, post-reversal checks); earlier commits for the validation tests

TABLE XVII. Main symbols.

T, σ	nominal tension, stress	$\Phi(\epsilon)$	strain energy per unit volume
$E_0, b = \beta/A$	initial modulus, cubic coefficient	e_{mat}	specific energy to failure
ϵ_b, ϵ_p	failure strain, prestrain	η	material utilization
$\mu_0 = \rho A$	linear density	Π_E, Π_T	energy and speed ratios, Eq. (6)
c_L, c_T	longitudinal, transverse wave speeds	S_N	junction sum, Eq. (15)
v_0, v_c	impact speed, critical speed	Z	wave impedance $\mu_0 c_L$
M, r_d	drone mass, radius	Λ_{loc}	localization number, Eq. (28)
N, R	number of radials, frame radius	R_d	radius of the damaged zone
q_e, L_q	force density, weighted Laplacian	w, w_b	hub deflection, deflection at failure

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